Fluids, Electrolytes, Acid-Base Disorders, and Nutrition Support

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Learning Objectives

1. Describe normal fluid requirements and common patient conditions that alter fluid needs and homeostasis.
2. Develop an appropriate assessment and treatment plan for common electrolyte disorders in critically ill patients.
3. Correctly identify complex acid-base disorders based on a patient's laboratory data and clinical course.
4. Specify the appropriate route (parenteral or enteral) of nutrition administration, energy and protein needs, and key micronutrients to be provided to a critically ill patient.
5. Develop an appropriate assessment of the tolerance, safety, and efficacy of an enteral or parenteral nutrition regimen.

Abbreviations in This Chapter

ABG	Arterial blood gas
ADH	Anti-diuretic hormone
AG	Anion gap
AKI	Acute kidney injury
APACHE	Acute physiology and chronic health evaluation (score)
BEE	Basal energy expenditure
BG	Blood glucose
BMI	Body mass index
BUN	Blood urea nitrogen
CKD	Chronic kidney disease
COPD	Chronic obstructive pulmonary disease
CRRT	Continuous renal replacement therapy
COVID-19	Coronavirus disease 2019
CSWS	Cerebral salt wasting syndrome
CVP	Central venous pressure
ECF	Extracellular fluid
EDVI	End-diastolic volume index
EN	Enteral nutrition
GRV	Gastric residual volume
HCO_3	Bicarbonate
IBW	Ideal body weight
ICF	Intracellular fluid
ICU	Intensive care unit
ILE	Lipid injectable emulsion
K	Potassium
NA	Sodium
NB	Nitrogen balance
NG	Nasogastric
NPO	Nil per os (nothing by mouth)
NRS	Nutrition risk screening (score)
NUTRIC	Nutrition Risk in the Critically Ill (score)
P_{CO_2}	Partial pressure of carbon dioxide
PCWP	Pulmonary capillary wedge pressure
PN	Parenteral nutrition
P_{O_2}	Partial pressure of oxygen
REE	Resting energy expenditure
SCR	Serum creatinine
SIAD	Syndrome of inappropriate antidiuresis
SOFA	Sequential organ failure assessment (score)
TBI	Traumatic brain injury
UUN	Urine urea nitrogen
WT	Body weight

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

1. Which is the most appropriate indication for parenteral nutrition (PN)?
   A. Severe anorexia.
   B. Lack of bowel sounds.
   C. Ileus.
   D. Gastric residual volume (GRV) of 300 mL.

2. Which fluid option would most closely match the composition of nasogastric (NG) fluid losses?
   A. 0.9% sodium chloride and potassium chloride 20 mEq/L.
   B. 0.45% sodium chloride and potassium chloride 20 mEq/L.
   C. 5% dextrose in 0.225% sodium chloride and potassium chloride 20 mEq/L.
   D. Lactated Ringer solution.

3. A patient with prolonged, intractable diarrhea would most likely have excess losses of which trace mineral?
   A. Zinc.
   B. Copper.
   C. Iodine.
   D. Manganese.

4. A 70-year-old man admitted to the intensive care unit (ICU) for sepsis was recently given a diagnosis of syndrome of inappropriate antidiuretic hormone secretion. His serum sodium fell from 130 mEq/L to 115 mEq/L during the past 3 days, and his course was subsequently complicated by a seizure. Which would be the most appropriate treatment option?
   A. Intravenous 0.9% sodium chloride.
   B. Intravenous desmopressin acetate (DDAVP).
   C. Intravenous 3% sodium chloride.
   D. Intravenous conivaptan.

5. Other than the absorption/infusion rate, which best explains why enteral potassium administration is safer than parenteral potassium administration?
   A. Bioavailability of potassium is significantly lower with enteral versus parenteral administration.
   B. Feed-forward sensing of changes in mesenteric potassium concentration increases urinary potassium excretion.
   C. Potassium chloride elixir is likely to cause diarrhea and reduce potassium absorption.
   D. Wax matrix tablets sequester potassium release throughout the gastrointestinal (GI) tract.

6. A 40-year-old man (weight 60 kg) is admitted to the trauma ICU after a motor vehicle collision. He is noted to have a serum magnesium concentration of 1.2 mg/dL, and his family states that he has a history of alcohol abuse (12–18 beers/day). He is given magnesium sulfate 6 g intravenously for 6 hours by the primary service. His repeat serum magnesium concentration on the following day is 1.8 mg/dL. Which would be the most appropriate treatment for this patient?
   A. No treatment is necessary because his serum magnesium concentration is normal.
   B. If a repeat serum magnesium concentration is 2 mg/dL or greater, no additional magnesium therapy is indicated.
   C. Supplemental calcium therapy should be given concurrently with the magnesium therapy.
   D. Additional magnesium therapy should be given daily for the next 4–5 days.

7. A 45-year-old man (weight 78 kg) with a history significant for hypertension and pancreatitis is admitted to the ICU after operative management of necrotizing pancreatitis. He is given PN consisting of 650 g of dextrose, 120 g of amino acids, and 55 g of lipid injectable emulsion (ILE) daily. Blood glucose (BG) measurements over the past 24 hours have been 180-325 mg/dL. Which change to this patient's nutrition regimen is most appropriate at this time?
   A. Initiate a sliding-scale insulin regimen with insulin aspart every 4 hours.
   B. Temporarily decrease the dextrose dose in the PN to 200 g and advance to dextrose 650 g once blood glucose is less than 180 mg/dL.
   C. Decrease dextrose to 500 g and make up with remaining kilocalories by increasing protein to 160 g.
   D. Decrease dextrose to 315 g and reassess glycemic control over the next 24 hours.

8. A 48-year-old man is admitted to the trauma ICU after a motorcycle collision. His injuries include a subarachnoid hemorrhage, right pneumothorax, multiple rib fractures, grade 5 liver laceration, right sacral fracture, and transverse process fractures. His course is complicated by respiratory failure, acute kidney injury (AKI), and hyperglycemia. His labs today reveal serum sodium 141 mEq/L, chloride 102 mEq/L, carbon dioxide 20 mEq/L, and lactate 2.6 mmol/L. His arterial blood gas values are as follows: pH 7.46, P_{CO_2} 31 mmHg, and HCO_3 22 mEq/L, Which of the following assessments of his acid-base status is correct?
   A. Primary respiratory alkalosis only.
   B. Primary respiratory alkalosis with underlying metabolic acidosis.
   C. Primary metabolic alkalosis with underlying respiratory alkalosis.
   D. Primary metabolic acidosis only.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 6, 8
   b. Task B: 5, 9, 13-15
   c. Task C: 1, 3, 4
3. Domain 3: Professional Practice
   a. Task A: 3
   b. Task B: 1, 7

I. FLUIDS AND ELECTROLYTES

A. General Overview

1. Body water compartments
   a. Total body water (TBW): About 60% of body weight for men; about 50% of body weight for women; lower percentage for those with obesity and for older adults (0.5 L/kg for men; 0.45 L/kg for women)
   b. About 60% of TBW is intracellular.
   c. About 40% of TBW is extracellular water (about 75% is interstitial fluid; about 25% plasma volume).
   d. Fluid compartments are separated by membranes that are freely permeable to water. The movement of fluids through these compartments is due to hydrostatic pressure or osmotic pressure.
2. Estimating daily fluid requirements
   a. Simple, weight-based estimation: 30–35 mL/kg (overestimates needs for patients with higher BMI; underestimates needs for patients with lower BMI)
   b. Holliday-Segar formula: 100 mL/kg for the first 10 kg, 50 mL/kg for the next 10 kg, and 20 mL/kg thereafter
   c. Increased insensible losses occur with diarrhea and fever (around 10%–15% for every degree Celsius greater than 37°C).
   d. Fluids received from other sources should be considered when estimating a patient's fluid needs to avoid fluid overload.

Table 1. Effect of Body Temperature on Insensible Fluid Losses (Surgery 1968;64:154-64)

Rectal Temperature (°C)	No. of Patients	Mean Fluid Loss (mL/m^2/day)
36.7–37.7	205	552
37.8–38.2	160	600
38.3–38.8	48	768
38.9–40	14	840

3. Estimating electrolyte requirements
   a. Approximate electrolyte concentrations in the extracellular and intracellular fluids (ECFs and ICFs) (Fluid, Electrolyte, and Acid-Base Disorders, Vol 1. New York: Churchill Livingstone, 1985:1-38)

Table 2. Electrolyte Concentrations in the ECF and ICF

Electrolyte	Extracellular Fluid (mEq/L)	Intracellular Fluid (mEq/L)
	(plasma)	(muscle)
Sodium	140	12
Potassium	4.5	160
Chloride	104	2
Bicarbonate	24	10
Calcium	5.0 (10 mg/dL)	—
Magnesium	1.5 (1.8 mg/dL)	34
Phosphate	2 (3.5 mg/dL)	54

b. "Normal" daily requirements

Table 3. "Normal" Daily Requirements (Hosp Pharm 2002;37:1336-42; Medicine (Baltimore) 1981;60:339-54; Crit Care Med 1995;23:1504-11; Ann Rev Med 1981;32:245-59; Ann Surg 1983:197:1-6)

Electrolyte	Daily Requirements
Sodium	50–150 mEq (1–2 mEq/kg)
Potassium	0.5–1.5 mEq/kg
Phosphorus	10–30 mmol
Magnesium	8–32 mEq
Calcium	10–15 mEq
Chloride[a]	80–120 mEq
Acetate[a]	80–120 mEq

[a]Depending on the acid-base status of the patient.

c. Approximate electrolyte content of GI secretions (in milliequivalents per liter) (About Surgery: A Clinical Approach. Philadelphia: Elsevier Health Sciences, 1996:5-17; Acta Chir Scand Suppl 1963:suppl 306:301-65)

Table 4. Electrolyte Content of GI Secretions

Fluid	Average Daily Volume (mL)	Sodium (mEq/L)	Potassium (mEq/L)	Chloride (mEq/L)	Bicarbonate (mEq/L)	Magnesium (mEq/L)
Stomach	1000–2000	60–90	10–15	100–130	—	0.9
Duodenum	400–600	140	5–10	90–120	80	—
Jejunum/Ileum	2000–2500	140	5–10	90–120	30–40	6–12
Colon	< 300	60	20–30	50	—	6–12
Pancreas	600–800	140	5–10	75	115	0.4
Bile	300–600	140	5–10	100	30	1.1

d. Electrolyte composition of common intravenous solutions (in milliequivalents per liter)

Table 5. Electrolyte Composition of Common Intravenous Solutions

Solutions	Sodium (mEq/L)	Potassium (mEq/L)	Chloride (mEq/L)	Bicarbonate (mEq/L)	Calcium (mEq/L)	Magnesium (mEq/L)	Osmolarity (mOsm/L)
5% dextrose in water	—	—	—	—	—	—	252
0.45% sodium chloride (one-half normal saline)	77	—	77	—	—	—	154
5% dextrose in 0.45% sodium chloride (5% dextrose in one-half normal saline)	77	—	77	—	—	—	406
Lactated Ringer solution	130	4	109	28	2.7	—	274

Table 5. Electrolyte Composition of Common Intravenous Solutions (*continued*)

Solutions	Sodium (mEq/L)	Potassium (mEq/L)	Chloride (mEq/L)	Bicarbonate (mEq/L)	Calcium (mEq/L)	Magnesium (mEq/L)	Osmolarity (mOsm/L)
Plasma-Lyte A/Normosol-R pH 7.4	140	5	98	27	—	3	294–295
0.9% sodium chloride (normal saline)	154	—	154	—	—	—	308
3% sodium chloride (hypertonic saline)	513	—	513	—	—	—	1026

B. Water and Sodium Disorders
   1. Dehydration: Signs and symptoms include decreased urine output (unless patient has glycosuria or diuretic therapy), increased blood urea nitrogen/serum creatinine ratio (BUN/SCr greater than 20), insufficient net fluid balance (from intake and output documentation records), increased serum sodium, poor skin turgor, dry mucous membranes, orthostatic hypotension, "contraction alkalosis," increased losses
      a. Fever
      b. Skin losses through sweating
      c. GI fluid losses (e.g., emesis, gastric suction, diarrhea, enterocutaneous fistula, surgical wounds, drains)
   2. Volume excess: Signs and symptoms include peripheral/sacral/pulmonary edema, anasarca, rapid weight gain, positive fluid balance
      a. Excessive fluid and/or sodium intake
      b. Impaired ability to excrete excess water and sodium (e.g., heart failure, cirrhosis with ascites, renal failure)
   3. Hyponatremia (see also the Neurocritical Care chapter for further reading on hyponatremia)
      a. Classic evaluation
         i. Exclude hyperglycemia, mannitol, and glycine for unmeasured effective osmoles (hypertonic hyponatremia).
            (a) Correct serum sodium for hyperglycemia (once the hyperglycemia is controlled, the serum sodium will rise).
            (b) For every 100-mg/dL increase in BG greater than 100 mg/dL, serum sodium will fall by about 2.4 mEq/L (or vice-versa).
            (c) Corrected serum Na = Measured serum Na + [(BG -100)/100 * 2.4] (Am J Med 1999;106:399-403).
         ii. Exclude factitious/pseudo-hyponatremia (isotonic hyponatremia): Arguably still possible during lipemia (triglycerides greater than 1000 mg/dL or hyperproteinemia (e.g., multiple myeloma) if the serum is diluted under the assumption that the serum contains 7% solid-phase particles before the assay (Intensive Care Med 2014;40:320-31).
         iii. Evaluate ECF volume (increased, normal, decreased): Evaluate patient for edema, fluid balance on fluid intake/output records, hemodynamic markers, chest radiography: Pulmonary infiltrates without pneumonia, enlarged heart or evidence of diseases with decreased urinary water/sodium excretion (e.g., AKI, congestive heart failure, cirrhosis with ascites).
         iv. Consider using urine sodium and osmolality for hypotonic (serum osmolality less than 280 mOsm/kg) hyponatremia in conjunction with volume status (hypovolemic, euvolemic, hypervolemic):

Table 6. Comparative Features of Hypotonic Hyponatremia (Am J Med 2013;126:S1-42)

ECF volume status	Hypervolemic		Euvolemic		Hypovolemic	
Physiologic Findings	Edema, large positive fluid balance, pulmonary infiltrates without pneumonia, enlarged heart, increased PCWP, EDVI		No evidence of edema, fluid equilibrium, no evidence of dehydration or fluid overload, normal hemodynamics		Poor skin turgor, dry mucous membranes, decreased urine output, concentrated urine, tachycardia	
Urine Osm (mOsm/kg)	> 100		< 100	> 100	> 450	
Urine Na (mEq/L)	< 20	>20	< 20	> 20	< 20	>20
Potential etiologies	CHF, Cirrhosis with ascites, Nephrotic syndrome	Renal failure	Psychogenic polydipsia, Excessive hypotonic fluid intake, Beer potomania	SIAD[a], Cortisol deficiency, Hypothyroidism, Drug-induced	Extra-renal losses (e.g., GI fluid losses), Third space losses	Diuretics, Adrenal insufficiency, Cerebral salt wasting, thiazide diuretics, Salt wasting nephropathy

[a]Common etiologies for SIAD (Am J Kidney Dis 2008;52:144-53; Hosp Pharm 2002;37:1336-42):
Central nervous system (CNS) disorders – Trauma, stroke, infection, brain tumors
Malignancy – Small cell carcinoma of the lung, pancreatic carcinoma, lymphoma, Hodgkin disease, sarcoma
Pulmonary infection, respiratory failure with positive pressure ventilation
Endocrine disorders – Pituitary tumor, hypothyroidism, adrenal insufficiency
Stress response (surgery, trauma, thermal injury, sepsis, pain)
Drugs

CHF = congestive heart failure; EDVI = end-diastolic volume index; GI = gastrointestinal; PCWP = pulmonary capillary wedge pressure; SIAD = syndrome of inappropriate antidiuresis.

Table 7. Drug-Induced SIAD (Am J Kidney Dis 2008;52:144-53)

Mechanism	Examples
Increased hypothalamic production of ADH	Amitriptyline, imipramine
	Fluoxetine, sertraline, paroxetine
	Thioridazine, trifluoperazine
	Haloperidol
	Carbamazepine, oxcarbazepine, valproic acid
	Vincristine, vinblastine, cisplatin, carboplatin, cyclophosphamide, ifosfamide
	Nicotine
	Bromocriptine
	Monamine oxidase inhibitors
Increased sensitivity to or exogenous administration of ADH	DDAVP, desmopressin
	Chlorpropamide
	Lamotrigine

Table 7. Drug-Induced SIAD (Am J Kidney Dis 2008;52:144-53) (*continued*)

Mechanism	Examples
Mixed or uncertain	Opiates
	Barbiturates
	Nonsteroidal anti-inflammatory agents
	Angiotensin-converting enzyme inhibitors

ADH = antidiuretic hormone.

v. If the patient has a TBI, it can be difficult to ascertain whether the patient has cerebral salt wasting syndrome (CSWS) or SIAD.

Table 8. Comparison of Features of Hyponatremia Caused by CSWS vs. SIAD (Hosp Pharm 2002;37:1336-42)

CSWS	SIAD
Decreased serum sodium	Decreased serum sodium
Decreased ECF	Normal or expanded ECF
Negative sodium balance	Variable sodium balance
CVP/PCWP/EDVI decreased	CVP/PCWP/EDVI normal or increased
Urine osmolality increased	Urine osmolality increased
Urine sodium increased	Urine sodium increased

CVP = central venous pressure; CSWS = cerebral salt wasting syndrome; ECF = extracellular fluid.

b. 2014 European Society of Endocrinology guidelines (Intensive Care Med 2014;40:320-31)
   i. Exclude hyperglycemia and other causes of non-hypotonic hyponatremia.
   ii. Evaluate urine sodium and osmolality (the guidelines suggest that these markers should be assessed before ECF volume because the latter is difficult to determine in the critically ill patient).
      (a) The European guidelines also differ because they recommend a urine sodium concentration of 30 mEq/L (instead of 20 mEq/L, as previously described) as the point of demarcation for differentiating the etiology for hyponatremia.
      (b) If urine osmolality is less than 100 mOsm/kg, the guidelines recommend accepting relative excess water intake as a cause of the hypotonic hyponatremia.
   iii. Assess ECF and arterial blood volume, diuretics, presence of kidney disease (Am J Med 2010;123:652-7).
c. Treatment of hyponatremia
   i. Typically reserved for acute (less than 48 hours) hyponatremia and/or severe symptoms (mental status changes, coma, or seizures, regardless of chronicity) (Am J Med 2013;126:S1-42). Immediate treatment with hypertonic saline therapy (3% NaCl) should be adjusted to achieve a serum Na concentration increase of 5 mEq/L in the first hour, and limit to a total of 10 mEq/L increase during the first 24 hours and additional 8 mEq/L during every 24 hour thereafter until serum sodium reaches 130 mEq/L (Intensive Care Med 2014;40:320-31). More rapid correction could lead to central pontine myelinolysis. Discontinue hypertonic saline when goal serum sodium change has been achieved or symptoms have improved (whichever occurs first). Estimate sodium deficit with the following equation: sodium deficit (mEq) = TBW x (140 − measured serum sodium); TBW is estimated as 0.5 L/kg for women and 0.6 L/kg for men.

(a) Hypertonic saline dose and administration – Intermittent bolus versus continuous infusion:
(1) Intermittent boluses: 3% NaCl 150 mL (or 2 mL/kg) intravenously over 20 minutes ×3 (Intensive Care Med 2014;40:320-31) or 100 mL intravenously over 10 minutes ×3 (Am J Med 2013;126:S1-42)
(2) Continuous infusion: 3% NaCl 20 mL/hour (JAMA Intern Med 2021;181:1-12; J Clin Endocrinol Metab 2019;104:3595-602) or 0.5–2 mL/kg/hour (Am J Med 2013;126:S1-42)
(3) Bolus and continuous infusion dosing are both safe and effective with appropriate monitoring to minimize overcorrection. However, boluses may be favored for ease of administration.
(b) Prevention of overcorrection with preemptive administration of desmopressin (DDAVP Clamp) (Am J Kidney Dis 2013;61:571-8)
(1) Desmopressin 1–4 mcg intravenously every 6–8 hours + 3% NaCl continuous infusion to achieve a serum sodium increase of 6 mEq/L/day
(2) Stop once serum sodium has reach 128 mEq/L
(3) Not studied with hypertonic saline boluses (evaluated with continuous infusion only)
(4) Only appropriate for severe hyponatremia and high risk of overcorrection
(5) Avoid in patients with heart failure, hepatic cirrhosis, or volume overload
(c) Treatment of overcorrection or relowering of serum sodium is recommended for patients with a baseline serum sodium of less than 120 mEq/L who exceed recommended rates of lowering in the first 24 hours (Am J Med 2013;126:S1-42).
(1) Desmopressin 2–4 mcg intravenously every 8 hours
(2) Free water replacement of 3 mL/kg/hour either enterally with water or parenterally with 5% dextrose in water
(3) Recheck serum sodium every hour and continue until serum sodium goal is achieved

ii. Hypervolemic (ECF expanded) – Fluid and sodium restriction, diuretic therapy as needed. Consider conivaptan or tolvaptan if other therapies fail.

iii. Hypovolemic (ECF reduced) and low urine sodium – Give sodium and fluids for volume expansion (treat etiologies if possible); reduce diuretic therapy. Sodium and fluid administration can be done using 0.9% sodium chloride, lactated Ringer solution, or PlasmaLyte/Normosol intravenously.

iv. Euvolemic (ECF normal) – Consider syndrome of inappropriate antidiuresis or secondary adrenal insufficiency – Fluid restriction first (if clinically appropriate); free water restriction with use of 0.9% sodium chloride solution with or without diuretic therapy. Consider hypertonic saline for acute or severe symptoms. Consider salt tablets, urea, or demeclocycline to maintain normal serum sodium concentrations. Consider conivaptan or tolvaptan if other measures are not effective.

4. Hypernatremia: Reflects a deficit of water compared with total body sodium. Therapy should be adjusted to achieve a serum sodium concentration decrease of 6–8 mEq/L per day; no faster than 10–12 mEq/L per day to reduce the risk of cerebral edema.
a. Evaluation
i. Evaluate ECF volume status (increased, normal, decreased). Increased ECF volume is caused by excessive sodium intake (oral salt tablets, hypertonic saline, 0.9% sodium chloride solution, lactated Ringer solution, PlasmaLyte/Normosol). This is the least common cause of hypernatremia. Decreased ECF (hypovolemia, dehydration) results from excessive losses of both water and sodium.
ii. Consider urine sodium together with volume status to determine cause and treatment.

b. Treatment
   i. Hypervolemic hypernatremia – Treated with diuretics, sodium restriction, water replacement, and/or renal replacement therapy
   ii. Euvolemic hypernatremia – Treated with free water replacement
      (a) For normal or increased ECF volume, estimate free water deficit with the Adrogue-Madias equation (N Engl J Med 2000;342:1493-9): 0.6 x Wt (kg) x [(serum sodium/140) – 1] (use 0.5 x Wt (kg) for women). Correct deficit over 2–3 days. The equation often underestimates TBW deficit and should only be used a gross estimate for water deficit (Am J Clin Nutr 2013;87:79-85). This is typically accomplished using enteral free water boluses (e.g., 200–300 mL every 4–6 hours) if the patient has a nasogastric feeding or suction tube. Administration of high volumes of enteral water boluses should be avoided if the patient has a post-pyloric tube because it causes cramping and diarrhea. Rare cases of bowel necrosis post–large water boluses have also occurred when administered directly into the small bowel (J Parenter Enteral Nutr 2004;28:27-9). If the enteral route is not possible or the patient is intolerant of enteral water boluses, intravenous dextrose 5% in water or dextrose 5% in 0.225% sodium chloride can be used.
      (b) The anticipated change in serum sodium concentration after 1 L of any intravenous fluid can be estimated as follows: [sodium concentration of IV fluid (mEq/L) – serum sodium concentration (mEq/L)] / [TBW + 1] (N Engl J Med 2000;342:1493-9). Refer to Table 5 for sodium concentration of commonly used intravenous fluids.
   iii. Hypovolemic hypernatremia – Initially treated with volume expansion to restore hemodynamic stability because total body sodium and volume is low. If hypernatremia is still present once the patient is hemodynamically stable, free water replacement can be initiated as stated earlier.
   iv. Clinical evaluation during volume repletion is of paramount importance. Serum sodium should be checked routinely (e.g., every 6–12 hours) to avoid overcorrection. All sources of excess fluid losses should be monitored and replaced with the appropriate fluid, if indicated.

Table 9. Comparative Features of Hypernatremia (Nutr Clin Pract 2008;23:108-21)

ECF Volume Status	**Hypervolemic**	**Euvolemic**	**Hypovolemic**	
Physiologic findings	See Table 6			
Urine sodium (mEq/L)	> 20	Varies	< 20	> 20
Potential etiologies	Excessive sodium intake from intravenous or oral, mineralocorticoid excess	Extrarenal losses (insensible losses), Renal losses (diabetes insipidus)	Extrarenal losses (fever, diarrhea, respiratory losses)	Renal losses (diuretics, glycosuria, kidney failure/injury), diabetic ketoacidosis, hyperosmolar hyperglycemic syndrome

Patient Case

Questions 1–3 pertain to the following case.
A 55-year-old woman (70 kg) admitted to the ICU for pneumonia and respiratory failure develops a serum sodium of 125 mEq/L on her fifth day of hospital admission. Her other laboratory values include a serum potassium of 4.6 mEq/L, chloride (Cl) 100 mEq/L, total carbon dioxide (CO_2) content 24 mEq/L, BUN 20 mg/dL, serum creatinine (SCr) 1.1 mg/dL, and glucose 167 mg/dL. She currently receives a 1-kcal/mL, 62 g of protein/L enteral feeding formula at 60 mL/hour and a 5% dextrose in 0.45% sodium chloride infusion at 25 mL/hour. Her fluid balance has ranged from +300 to +600 mL/day during the past 3 days. She has no evidence of any significant amount of edema. Her measured serum osmolality is 265 mOsm/kg, urine osmolality is 490 mOsm/kg, and urine sodium is 67 mEq/L.

1. Which is the most likely etiology for the patient's hyponatremia?
   A. Factitious hyponatremia
   B. Adrenal insufficiency
   C. Cerebral salt wasting
   D. SIAD

2. Which would be the most appropriate treatment for this woman?
   A. Give sodium chloride tablets 1 g three times daily.
   B. Limit fluids.
   C. Change the intravenous fluid to 0.9% sodium chloride.
   D. Provide a short-term intravenous infusion of 3% sodium chloride.

3. Which change in the enteral feeding formula would be best for this patient?
   A. Add sodium chloride 100 mEq/L to the current formula.
   B. Change the formula to a fish oil–enriched product.
   C. Change the formula to a low-carbohydrate, high-fat product.
   D. Change the formula to a 2-kcal/mL formula, and decrease the rate.

C. Disorders of Potassium Homeostasis
   1. Potassium homeostasis overview
      a. 98% intracellular
      b. Total body stores: 35–50 mEq/kg in normal healthy adults; 25–30 mEq/kg if significantly undernourished
      c. Normal serum concentration: 3.5–5.2 mEq/L
      d. Serum concentration can be influenced by changes in pH (for every 0.1 increase in arterial pH, serum potassium will decrease by around 0.6 mEq/L [range 0.4–1.3 mEq/L]) (J Clin Invest 1956;35:935-9), and vice versa. This occurs because of potassium exchanging with hydrogen by the H+/K+-ATPase pump.
      e. Average daily requirement: About 0.5–1.5 mEq/kg
      f. Kidney is primary route of elimination.
      g. Losses can be extensive with severe diarrhea or body fluid drainages (see Table 4).
      h. Magnesium status can influence potassium homeostasis (J Am Soc Nephrol 2007;18:2649-52; Crit Care Med 1996;24:38-45; Arch Intern Med 1992;152:40-5).
         i. Magnesium serves as a cofactor for the Na-K-ATPase pump.
         ii. Magnesium closes potassium channels in distal nephron.

2. Hypokalemia
   a. Definition: Serum potassium less than 3.5 mEq/L, though most ICUs empirically prefer to keep patients at 4.0 mEq/L or greater, if possible.
   b. Signs and symptoms: Weakness, cramps, cardiac arrhythmias (ST depression, QT prolongation, flat T wave, U wave). If severe hypokalemia (e.g., serum potassium less than 2 mEq/L): flaccid paralysis, ileus.
   c. Etiologies:
      i. Inadequate intake (rare; kidneys can usually adapt)
      ii. Increased losses
         (a) GI fluid losses (e.g., diarrhea, fistula, drainages)
         (b) Hypomagnesemia
         (c) Medications (diuretics, amphotericin B, mineralocorticoid excess, cisplatin, extended-spectrum penicillins such as piperacillin, ticarcillin)
         (d) Polyuria (diabetes insipidus)
         (e) Renal potassium excretion (type I/distal and type II/proximal renal tubular acidosis)
         (f) Diabetic ketoacidosis
      iii. Increased requirements (building of new muscle/tissue – refeeding syndrome)
      iv. Extracellular to intracellular shift
         (a) Medications (β-adrenergic agonists, including albuterol, sodium bicarbonate or other alkalinizing agents; insulin)
         (b) Acute alkalemia
         (c) Hypothermia
         (d) Pentobarbital
   d. Treatment:
      i. Treat, alleviate, or reduce the potential etiologies for hypokalemia, if possible.
      ii. Ensure that hypokalemia is not at least partly attributable to hypomagnesemia.
      iii. The estimated deficit should be replaced during a period of 1–3 days (depending on the extent of deficit; the larger the deficit, the longer the repletion period) by giving boluses and increasing the potassium content in intravenous fluids and/or PN formulation.
      iv. Enteral or oral potassium replacement is the preferred and safer route of delivery in asymptomatic patients because of the time of absorption and feed-forward regulation of potassium homeostasis (Ann Intern Med 2009;150:619-25); oral potassium is available in an extended release tablet or capsule (potassium chloride), oral solution (potassium chloride), and effervescent tablet (potassium bicarbonate); the oral solution is limited by poor palatability (preferred for administration via nasogastric/orogastric tube), and bicarbonate salt forms may not be appropriate for patients with a significant metabolic alkalemia; administration of potassium chloride liquid directly into the small bowel (by a jejunal or duodenal feeding tube) should be avoided because of its osmolality, which can lead to abdominal cramping, distension, and diarrhea.
      v. Central administration of intravenous repletion doses of potassium chloride or potassium phosphate is preferred. Potassium chloride can be given at 20 mEq/hour if the patient has continuous electrocardiography (ECG) monitoring. Ten mEq/hour is safest if the patient is asymptomatic. Peripheral intravenous solutions should not contain potassium chloride at more than 40–60 mEq/L in an effort to reduce the pain associated with the infusion of a concentrated potassium chloride solution and to prevent inappropriate rapid and excessive potassium chloride dosing.
      vi. Empiric intravenous potassium dosing. Institution or ICU-specific repletion protocols are common and may need to be adjusted according to patient body size, renal function, ongoing losses, and response to previous boluses. The following algorithm is an example of empiric potassium repletion that is based on serum concentrations.

Table 10. Empiric Intravenous Potassium Dosing

Serum Potassium (mEq/L)	Potassium Chloride Dosage (mEq)[a]	Laboratory Tests
3.5–3.9	40 to 60 mEq	Obtain BMP, magnesium next AM
3–3.4	80 mEq	Obtain BMP, magnesium next AM; may obtain potassium 1–2 hours after repletion is completed, especially if losses are thought to be high; reassess
2–2.9	120 mEq	Obtain repeat serum potassium 1–2 hours after repletion is completed and reassess; may need one or two additional boluses; repeat; check serum magnesium next AM; reassess

[a]Potassium phosphate may be considered in lieu of potassium chloride if concurrent hypokalemia and hypophosphatemia (very common in critically ill patients receiving EN/PN). Thirty millimoles of potassium phosphate is equivalent to 44 mEq of potassium.

AM = morning; BMP = basic metabolic panel.

vii. The historical assumption of "a 0.5 to 0.6 mEq/L increase in serum potassium will occur for every 40 mEq of intravenous potassium administered" (J Clin Pharmacol 1994;34:1077-82; Arch Intern Med 1990;150:613-7) is potentially inaccurate for many critically ill subpopulations such as emaciated patients or patients with obesity, those with renal dysfunction, exaggerated requirements such as trauma or thermally injured patients (J Parenter Enteral Nutr 2017;41:796-804), those with volume overload, or those receiving diuretic therapy.

viii. Serum potassium concentrations are equilibrated within 1–2 hours after completion of the intravenous potassium chloride infusion (J Clin Pharmacol 1994;34:1077-82; Crit Care Med 1991;19:694-9), and repeated assessments are recommended for patients with severe and/or complicated cases of hypokalemia.

3. Hyperkalemia
   a. Definition: Serum potassium greater than 5.2 mEq/L, although usually not a significant problem until serum potassium approaches 6 mEq/L
      i. Rule out factitious hyperkalemia (hemolysis of blood sample, white blood cell count greater than 10 x 10^3 cells/mm^3, platelet count greater than 400,000/mm^3). The potential for this error can be reduced by collecting the blood sample in a heparinized tube. Another source of factitious hyperkalemia occurs when the blood sample is obtained from the same intravenous line as a potassium-containing fluid.
      ii. Assess arterial blood gas (ABG) (severe acidosis).
      iii. Assess for recent administration of blood products (red blood cells contain 7.5–13.5 mEq/L of potassium).
   b. Signs and symptoms: ECG changes (peaked and tented T waves) and arrhythmias (bradyarrhythmias, ventricular fibrillation, asystole), symptoms similar to those of hypokalemia (weakness, paralysis)
   c. Etiologies:
      i. Drugs – Potassium-sparing diuretics (spironolactone, amiloride, triamterene), angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, nonsteroidal anti-inflammatory drugs, heparin, trimethoprim, octreotide, succinylcholine, digoxin (toxicity)
      ii. Excessive intake (usually in combination with compromised renal function) – Be sure to examine all intravenous fluids, EN and PN regimens, penicillin G (1.7 mEq of potassium per million units), packed red blood cells.
      iii. Renal dysfunction (chronic kidney disease [CKD], AKI)
      iv. Hyporeninemic hypoaldosteronism
      v. Tissue catabolism (chemotherapy, rhabdomyolysis, tumor lysis syndrome)

vi. Severe acidemia
vii. Older adult patients are also at risk because of decreased renal function, reduced renal functional reserve, and loss of body cell mass.

d. Treatment
i. The following regimens are for patients with symptomatic evidence of toxicity from hyperkalemia such as peaked T waves on ECG:

Table 11. Treatment of Hyperkalemia

Mechanism	Treatment	Description
Stabilization of myocardium	Calcium gluconate 10%, 1–2 g IV	Given as slow intravenous push (especially if ECG changes). Repeat dose after 5 minutes if ECG changes persist. Should be given in addition to other therapies as calcium does not affect serum potassium concentration.
	Calcium chloride, 0.5–1 g IV	
Intracellular shifting of potassium (temporary redistribution)	Regular human insulin, 5–10 units IV	Usually given with dextrose 50 g IV. Dextrose can be omitted if the patient has hyperglycemia.
	Sodium bicarbonate, 50–100 mEq IV	Especially useful if patient is acidemic
	Albuterol, 10–20 mg via nebulization over 10 minutes	
Elimination of potassium	Loop diuretics	
	Sodium zirconium cyclosilicate (SZC)	See chapter text for details.
	Hemodialysis	The most effective and predictable method for potassium elimination but also the most invasive. Reserved for life-threatening hyperkalemia.

ECG = electrocardiogram.

ii. Ensure no exogenous sources of potassium are present (e.g., intravenous fluids, EN, PN); to reduce intake with EN or PN, use a "renal" (no or low-electrolyte) formulation, if necessary.
iii. Potassium binding agents may be considered in conjunction with standard of care for treatment of acute hyperkalemia (Mayo Clin Proc 2021;96:744-62). Compared with other potassium binders, sodium zirconium cyclosilicate (SZC) has higher selectivity for potassium and the quickest onset of action at 1 hour, which makes it the most favorable option in the acute setting. SZC and patiromer are both more palatable and have superior safety profiles compared with sodium polystyrene sulfate (SPS). The onset of SPS is variable (up to days), as is its efficacy in the acute setting. Additionally, the risk of rare but significant adverse events with the use of SPS, including bowel ischemia, bowel necrosis, and increased hospitalizations, limits its utility as adjunctive therapy for acute hyperkalemia.

D. Disorders of Magnesium Homeostasis
1. Magnesium homeostasis overview
a. 99% intracellular (17% of total body content is in the muscle or in the skeleton)
b. Total body stores: Around 2000 mEq
c. Normal serum concentration: 1.8–2.4 mg/dL (about 30% bound to protein)
d. Average daily requirement: Around 24–32 mEq/day

e. Kidney is primary route of elimination (around 70% reabsorbed in ascending loop of Henle) and is without any hormonal regulation of renal magnesium reabsorption.
f. Losses can be extensive with severe diarrhea or body fluid drainages (see Table 4).
g. Magnesium depletion can influence potassium and calcium homeostasis.

2. Hypomagnesemia
   a. Definition: Although the lower limit of normal for serum magnesium concentrations is 1.8 mg/dL (1.5 mEq/L), most clinicians define significant hypomagnesemia as 1.5 mg/dL (1.3 mEq/L) or less (Annu Rev Med 1981;32:245-59; Nutrition 1997;13:303-8). Many ICUs have a target serum magnesium concentration greater than 2 mg/dL (1.8 mEq/L). Serum concentrations of magnesium may be slightly falsely lowered in the presence of significant hypoalbuminemia (Annu Rev Med 1981;32:245-59; Nutrition 1997;13:303-8).
   b. Signs and symptoms: Muscle weakness, cramping, paresthesias, Chvostek and Trousseau signs, tetany, QT prolongation, hypokalemia, hypocalcemia
   c. Etiologies:
      i. GI losses (especially diarrhea) – Average stool loss of about 6 mEq/L; up to 10–12 mEq/L or greater for secretory diarrheal losses
      ii. Alcohol (increased renal excretion; impaired absorption; poor nutritional status of patients who abuse alcohol)
      iii. Sepsis/critical illness (increased urinary excretion – several factors)
      iv. Pancreatitis (partly attributable to calcium-magnesium soap formation in peritoneum)
      v. Thermal injury/TBI (increased urinary excretion – several factors)
      vi. Drugs – Diuretics, amphotericin B, caspofungin, cyclosporin/tacrolimus, foscarnet, pentamidine, piperacillin/tazobactam, cisplatin/carboplatin/ifosfamide/cetuximab, lactulose/orlistat, aminoglycosides, and potentially long-term use of digoxin or proton pump inhibitors
      vii. Polyuria (osmotic agents, hypercalcemia, ureagenesis)
   d. Estimating magnesium deficit: For a serum magnesium concentration of less than 1.5 mg/dL (1.3 mEq/L), a 1- to 2-mEq/kg deficit can be expected.
   e. Treatment:
      i. Treat the etiology (if possible). Be sure to treat magnesium deficiency at the same time or before potassium therapy if the patient is also hypokalemic.
      ii. Successful treatment of hypomagnesemia usually takes 4–5 days of intravenous therapy. Intramuscular magnesium therapy for replacement therapy is inadvisable given the limit on volume per injection site with respect to dosage requirements and tissue irritation.
      iii. Intravenous magnesium sulfate 32–48 mEq/day (4–6 g/day) – Suggested to be sufficient to maintain serum magnesium within 2–2.5 mg/dL for most magnesium-deficient patients (Crit Care Med 1996;24:38-45; Annu Rev Med 1981;32:245-59)
      iv. Empiric intravenous magnesium sulfate dosing. This algorithm (Table 12) was designed primarily for trauma and thermally injured patients but can likely be universally applied to other critically ill patient populations (Nutrition 1997;13:303-8). The therapy may need to be adjusted according to renal function and ongoing severity of losses. An empiric approach to dosing intracellular electrolytes for patients with significant renal impairment is to give one-half the recommended dose. However, electrolyte therapy for patients with renal impairment must be individualized according to patient response.

v. Magnesium concentrations are often elevated for several hours or longer after an infusion because it takes about 48 hours for the magnesium to fully redistribute to the body tissues (Nutrition 1997;13:303-8). One-time repletion doses are usually not adequate for large deficits, and levels should continue to be monitored daily to ensure an appropriate assessment of magnesium is made after redistribution of previous bolus doses.

Table 12. Empiric Intravenous Magnesium Sulfate Dosing

Serum Magnesium (mg/dL)	Dose (g/kg)[a]
1.6–1.8	0.05
1–1.5	0.1
< 1	0.15

[a]For ease of use and preparation, intravenous magnesium sulfate should be ordered in 2-g increments (e.g., 2 g, 4 g, 6 g). The drug should be mixed in 100–250 mL of normal saline or 5% dextrose and given at a rate no faster than 1 g (8 mEq) per hour (Nutrition 1997;13:303-8). Commercially available sources of magnesium sulfate are available in 16 mEq/50 mL and 32 mEq/100 mL premixed bags. Maximum dose should be held at a ceiling of 8–10 g per administration. Intravenous magnesium may be administered by the peripheral or central venous route.

vi. Oral magnesium: It can be difficult to successfully replenish magnesium if given by the oral route in critically ill patients because of the adverse GI effects of oral magnesium (e.g., diarrhea) and the high elemental magnesium doses required to achieve repletion. Although it has been inferred that certain oral magnesium products are better tolerated than others (e.g., gluconate vs. oxide), this tolerability likely pertains to the elemental magnesium content of the products. The lower the elemental magnesium content, the more tolerable the oral product. However, the lower the magnesium content, the more difficult it is to achieve magnesium repletion for a patient with significant magnesium depletion.

Table 13. Common Oral Magnesium Products

Salt Form	Strength (mg)	Elemental Magnesium Content (mEq)	Usual Dosing
Oxide	400 140	19.8 6.9	1–2 tablets two or three times daily
Gluconate	500	2.2	1–2 tablets two or three times daily
Chloride	520	2.6	1–2 tablets two or three times daily

Table 14. Oral Magnesium Content

Salt Form	% Elemental Magnesium Content
Oxide	60
Carbonate	45
Hydroxide	42
Citrate	16
Lactate	12
Chloride	12
Sulfate	10
Gluconate	5

3. Hypermagnesemia
   a. Definition: Serum magnesium concentration greater than 2.4 mg/dL
   b. Signs and symptoms: Hypotension, decreased deep tendon reflexes, cardiovascular manifestations (e.g., bradycardia, somnolence, muscle paralysis, arrhythmias) generally do not occur until serum concentrations are greater than 4 mg/dL.
   c. Etiologies: Renal failure or impairment, early post-infusion elevation of serum magnesium concentration, excessive dosing of magnesium/antacids, post-cathartic use (e.g., magnesium citrate) – To develop hypermagnesemia, these events usually occur together with renal impairment.
   d. Treatment:
      i. Remove source of magnesium intake.
      ii. Intermittent slow bolus doses of calcium gluconate (2 g) for 5–10 minutes until severe symptoms abate (the effect of calcium is transient, and repeat therapy may be needed as often as every hour). The effects of magnesium on neuromuscular and cardiac function are antagonized by calcium.
      iii. Ventilate the patient, if necessary.
      iv. 0.9% sodium chloride infusion with loop diuretic therapy
      v. Hemodialysis

Patient Case

Questions 4 and 5 pertain to the following case.
A 65-year-old man (weight 87 kg) is admitted to the hospital with severe acute pancreatitis. A computed tomography (CT) scan of the abdomen reveals a pancreatic pseudocyst. He is to remain NPO (nothing by mouth) because of marked abdominal pain on a low-fat diet and during a trial of EN; therefore, he is given PN. He is a 2 pack/day tobacco smoker and has a history of frequent alcohol consumption. His serum laboratory values are as follows: sodium (Na) 139 mEq/L, potassium (K) 3.3 mEq/L, Cl 102 mEq/L, total CO_2 content 25 mEq/L, BUN 14 mg/dL, SCr 0.8 mg/dL, calcium 7.6 mg/dL, phosphorus 2.2 mg/dL, magnesium 1.5 mg/dL, and albumin 2.5 g/dL.

4. Which potassium-phosphorus dosing regimen would be most appropriate for this patient?
   A. Potassium phosphate 30 mmol intravenously x 1 dose, followed by potassium chloride 40 mEq via NG tube x 2 doses.
   B. Potassium phosphate 30 mmol intravenously x 1 dose, followed by potassium chloride 40 mEq intravenously x 1 dose.
   C. Potassium phosphate 60 mmol intravenously x 1 dose.
   D. Potassium chloride 40 mEq via NG tube x 2 doses, followed by Neutra-Phos 250 mg via NG tube x 2 doses.

5. In addition to potassium and phosphorus supplementation, the patient is given magnesium sulfate 6 g intravenously for 6 hours. His repeat serum magnesium the next day is 2.0 mg/dL. Which therapeutic option would be best for this patient?
   A. Give magnesium oxide 500 mg twice daily for the next few days.
   B. Give magnesium sulfate 2–4 g intravenously daily for the next few days.
   C. Give an additional dose of 8 g of magnesium sulfate intravenously.
   D. No additional treatment is necessary.

E. Disorders of Calcium Homeostasis
   1. Calcium homeostasis overview
      a. Most prevalent intracellular cation in body; 99% of body's calcium is in bone; highly protein bound in plasma
      b. Total body stores: About 1–1.2 kg of calcium
      c. Normal serum concentration: 8.5–10.5 mg/dL; normal serum ionized concentration 1.12–1.32 mmol/L
      d. Serum concentration can be influenced by:
         i. Changes in plasma albumin concentration – For every 1 g/dL in serum albumin below 4 g/dL, serum calcium will decrease by around 0.8 mg/dL (Clin Chim Acta 1971;35:483-9). This estimation of serum calcium concentrations is inaccurate in critically ill patients and should not be used (JPEN J Parenter Enteral Nutr 2004;28:133-41). Ionized calcium concentrations should be used for assessing calcium in critically ill patients. However, most critically ill patients (85%) with a total serum calcium concentration less than 7 mg/dL are hypocalcemic (ionized serum calcium of 1.12 mmol/L or less) (Nutr Clin Pract 2007;22:323-8).
         ii. Changes in pH (for every 0.1-unit increase in arterial pH, serum ionized calcium will decrease by about 0.05 mmol/L) (Arch Pathol Lab Med 2002;126:947-50) because of increased protein binding
      e. Average daily requirement: 10–15 mEq/day intravenously with PN (Ann Surg 1983;197:1-6); 1–3 g/day orally
      f. Kidney is primary route of elimination.
      g. Magnesium status can influence calcium homeostasis (J Parenter Enteral Nutr 2004;28:133-41; Annu Rev Med 1981;32:245-59).
         i. Hypomagnesemia results in end-organ resistance to parathyroid hormone.
         ii. Hypomagnesemia may impair parathyroid hormone secretion.
         iii. Hypocalcemia will correct within 2 days after hypomagnesemia is corrected.
   2. Hypocalcemia
      a. Definition: Corrected serum total calcium less than 8.5 mg/dL (non-ICU patients); ionized serum calcium concentration less than 1.12 mmol/L
      b. Signs and symptoms: Tingling, paresthesias, hyperactive deep tendon reflexes, tetany (Chvostek and Trousseau signs), seizures, prolonged QT interval
      c. Etiologies
         i. Critical illness, surgery
         ii. Continuous renal replacement therapy (CRRT) and citrate anticoagulation
         iii. Massive blood transfusion
         iv. Hypomagnesemia
         v. Hyperphosphatemia
         vi. Pancreatitis
         vii. Drugs (amphotericin B, cisplatin, cyclosporine, foscarnet, bisphosphonates, loop diuretics, sodium bicarbonate, cinacalcet, fluoride poisoning)
         viii. Malabsorption
         ix. Hypoparathyroidism, hypothyroidism
         x. Chronic kidney injury/AKI
         xi. Vitamin D deficiency
         xii. Severe alkalemia
      d. Treatment: See Table 15 for an empiric algorithm for intravenous repletion of calcium. (JPEN J Parenter Enteral Nutr 2007;31:228-33; JPEN J Parenter Enteral Nutr 2005;29:436-4; Nutrition 2007;23:9-15).

Table 15. Empiric IV Calcium Gluconate Dosing

Ionized Calcium (mmol/L)	Intravenous Calcium Gluconate Dosage (g)[a,b,c]
1–1.12	2 g (9.3 mEq) of calcium gluconate in 100 mL of 0.9% NaCl or D_5W for 2 hours
≤ 0.99	4 g (18.6 mEq) of calcium gluconate in 100 or 250 mL of 0.9% NaCl or D_5W for 4 hours

[a]For ease of use and preparation, calcium gluconate should be ordered in gram increments. Calcium chloride should be used (preferably) only in code situations, not for routine replacement therapy, because the chloride salt contains about 2.5 times the amount of elemental calcium and can cause tissue necrosis when given peripherally in contrast to calcium gluconate. However, during extreme circumstances, such as a national drug shortage of intravenous calcium gluconate, calcium chloride can be given in 0.67- and 1.3-g doses in lieu of 2- and 4-g calcium gluconate doses. In addition, calcium chloride should never be added to the PN solution unless there is no phosphate in the PN solution and the commercial amino acids used in the PN solution do not contain phosphorus (some amino acid products do contain phosphate). A serum ionized calcium concentration determination should be repeated several hours after completing the calcium gluconate infusion to allow equilibration (Nutrition 2007;23:9-15). More aggressive therapy may need to be considered for patients with tetany or life-threatening cardiac arrhythmias caused by hypocalcemia.

[b]Intravenous calcium administration should be used with extreme caution in patients with severe hypokalemia or in those receiving digoxin or other digitalis alkaloids.

[c]Always check the serum phosphorus concentration because hyperphosphatemia can induce hypocalcemia, given the metastatic precipitation of calcium phosphate in the soft tissues and lungs (usually associated with renal disease).

D5W = 5% dextrose in water; NaCl = sodium chloride.

Patient Case

Questions 6 and 7 pertain to the following case.
A 24-year-old man (weight 90 kg) is admitted to the trauma ICU postoperatively from repair of his duodenal, jejunal, ileal, and colon injuries; hepatorrhaphy; and splenectomy after several gunshot wounds to the abdomen. He also received 10 units of packed red blood cells. He has a serum ionized calcium concentration of 0.86 mmol/L, K 4.6 mEq/L, and magnesium 1.8 mg/dL. His SCr concentration is 0.8 mg/dL, and his urine output is 0.5 mL/kg/hour.

6. Which is the most likely etiology of his hypocalcemia?
   A. Hypomagnesemia
   B. Excessive urinary diuresis
   C. Blood transfusion
   D. Critical illness

7. Which therapeutic regimen would be best for this patient?
   A. Calcium gluconate 2 g intravenously for 2 hours
   B. Calcium gluconate 4 g intravenously for 4 hours
   C. Calcium chloride 1 g intravenous push for 5–10 minutes
   D. No calcium therapy necessary.

3. Hypercalcemia
   a. Definition: Corrected serum calcium greater than 10.5 mg/dL or ionized calcium greater than 1.32 mmol/L; signs and symptoms are more evident when total serum calcium of 12 mg/dL or greater or ionized calcium of 1.5 mmol/L or greater.
   b. Signs and symptoms: Mental status changes, polyuria, shortened QT interval, bradycardia, atrioventricular block
   c. Etiologies:
      i. Immobilization
      ii. Chronic critical illness–associated metabolic bone disease
      iii. Excessive calcium or vitamin D intake
      iv. Hyperparathyroidism

v. Granulomatous diseases (tuberculosis, sarcoidosis)
vi. Malignancy
vii. Drugs (thiazide diuretics, vitamin D, lithium, teriparatide)
viii. Dehydration
ix. Thyrotoxicosis
x. Adrenal insufficiency

d. Treatment:
i. Mobilize the patient (if possible); discontinue calcium supplementation
ii. Intravenous fluids with 0.9% sodium chloride (if dehydrated) at 200–300 mL/hour x 48 hours or until rehydrated with or without furosemide 40–80 mg intravenously every 12 hours
iii. Calcitonin 4 units/kg intramuscularly or subcutaneously every 12 hours; can be increased to a maximum of 8 units/kg every 8 hours. Calcitonin inhibits bone reabsorption and promotes the renal excretion of calcium by decreasing tubular reabsorption. The onset of action is rapid (within hours); however, tachyphylaxis can occur because of down-regulation of calcitonin receptors within days (J Intensive Care Med 2015;30:235-52) so duration of use should be limited to 24–72 hours. Calcitonin is most useful in patients with symptomatic hypercalcemia in combination with rehydration.
iv. Bisphosphonates. Pamidronate 90 mg intravenously once for acute hypercalcemia unrelated to causes c. i. and c. ii. Pamidronate 30 mg intravenously daily for 3 days may be used if hypercalcemia in chronic critical illness is associated with metabolic bone disease or immobilization (Chest 2000;118:761-6). Zoledronic acid may also be an option; however, evidence regarding its efficacy for chronic critical illness–associated metabolic bone disease is lacking. Onset of action for bisphosphonates in lowering serum calcium is typically after 48 hours. Bisphosphonates should be used with caution in patients with renal impairment.
v. Denosumab is preferred in patients with bisphosphonate-refractory hypercalcemia or those with poor kidney function (JAMA 2022;328:1624-36). Although it is not renally excreted, the degree of kidney impairment that warrants the use of denosumab over other options for hypercalcemia is not clearly established. Further, the presence of kidney dysfunction increases the risk of hypocalcemia from denosumab; therefore, a single dose of 60 mg or 0.3 mg/kg subcutaneously may be used in patients with kidney disease or in bishphosphonate-naive patients with moderate hypercalcemia. Otherwise, doses up to 120 mg once weekly for 3 weeks can be used. If the etiology of hypercalcemia persists, then 120 mg can continue to be administered every 4 weeks starting 2 weeks after the initial 3 weekly doses.
vi. Parathyroidectomy for patients with primary hyperparathyroidism. If surgery is not an option, cinacalcet may be used for patients with severe, chronic hypercalcemia from hyperparathyroidism (JAMA 2022;328:1624-36).
vii. Prednisone 40 mg/day and greater for 10 days for patients with granulomatous diseases (e.g., sarcoidosis, tuberculosis)
viii. Hemodialysis may be necessary for severe hypercalcemia.

F. Disorders of Phosphorus Homeostasis
1. Phosphorus homeostasis overview
a. 99% intracellular, of which 85% is bound to bone
b. Extracellular pool of phosphorus: Around 600 mg (about 20 mmol), 10% protein bound
c. Normal serum concentration: 2.5–4.5 mg/dL
d. Serum concentration can be influenced by parathyroid hormone (increased parathyroid hormone leads to increased urinary excretion of phosphorus), and alkalemia can decrease serum phosphorus concentration.

e. Average daily requirement: Around 20 mg/kg/day
f. Kidney is primary route of elimination.

2. Hypophosphatemia
   a. Definition: Serum phosphorus less than 2.5–3 mg/dL; severe hypophosphatemia less than 1–1.5 mg/dL
   b. Signs and symptoms: Weakness, paresthesias; severe depletion can lead to congestive cardiomyopathy, cardiac arrest, seizures, coma, respiratory arrest due to weakness of the diaphragm, rhabdomyolysis
   c. Etiologies:
      i. Alcoholism
      ii. Malnutrition/refeeding syndrome
      iii. Critical illness (especially trauma, TBI, thermal injury)
      iv. Diabetic ketoacidosis
      v. Hepatic resection
      vi. Drugs – Insulin, catecholamines, antacids, sucralfate, calcium, ferric carboxymaltose injection, diuretics
      vii. Alkalemia
      viii. Malabsorption – Chronic diarrhea
      ix. Hyperparathyroidism
      x. Cancer (phosphatonins [e.g., fibroblast growth factor-23])
      xi. Renal replacement therapies
      xii. Hungry bone syndrome following parathyroidectomy or thyroidectomy
   d. Treatment
      i. Target serum phosphorus concentration when patient is in the ICU and ventilator dependent: About 4 mg/dL (Intensive Care Med 1995;21:826-31; N Engl J Med 1985;313:420-4)
      ii. Intravenous dosing guidelines

Table 16. Intravenous Phosphorus Dosing Guidelines[a,b]

Serum Phosphorus (mg/dL)	General Medical-Surgical Population Dosage (mmol/kg) (Crit Care Med 1995;23:1504-11)	High Requirements[c] Population Dosage (mmol/kg) (JPEN J Parenter Enteral Nutr 2006;30:209-14)
2.3–3	0.16	0.32
1.6–2.2	0.32	0.64
< 1.6	0.64	1

[a]The drug should be mixed in 100–250 mL of normal saline or 5% dextrose in water and given at a rate no faster than 7.5 mmol/hour. Phosphorus should always be ordered in millimoles for ease of use and preparation. For ease of use and preparation in the pharmacy, phosphorus should be ordered in units divisible by 3 mmol (e.g., 15, 30, 45, 60) whenever possible.

[b]Potassium phosphate salt can be used for patients with a serum potassium less than 4 mEq/L, but ensure potassium content does not exceed estimated potassium repletion needs (for most products, 3 mmol of phosphorus = 4.4 mEq of potassium; for one commercially-available product, 3 mmol of phosphorus = 4.7 mEq of potassium). Sodium phosphate salt should be used for patients with a serum potassium of 4 mEq/L or greater or if the potassium content of potassium phosphate dose would exceed potassium repletion needs (3 mmol of phosphorus = 4 mEq of sodium).

[c]Patients with thermal injury (JPEN J Parenter Enteral Nutr 2001;25:152-9), those with trauma (especially those with a traumatic brain injury) (Nutrition 2010;26:784-90; JPEN J Parenter Enteral Nutr 2006;30:209-14), those malnourished with evidence of significant complications from refeeding syndrome, or those with hepatic resections.

iii. Repletion using oral phosphorus formulations
(a) Difficult to accomplish because of erratic absorption and adverse effects (e.g., diarrhea). This is especially true for severe hypophosphatemia in which higher doses are required.
(b) Neutra-Phos and Neutra-Phos K (only 8 mmol of phosphorus per tablet/packet)

3. Hyperphosphatemia
   a. Definition: Serum phosphorus concentration greater than 5 mg/dL, usually not clinically relevant until serum phosphorus is greater than 6 mg/dL
   b. Signs and symptoms: Hypocalcemia and metastatic calcification (e.g., neuromuscular irritability, prolonged QT interval, tetany) – Usually does not occur until the serum calcium-phosphorus product approaches 55 mg/dL or greater (Adv Exp Med Biol 1978;103:195-201).
   c. Etiologies:
      i. Renal failure
      ii. Immobility
      iii. Chronic critical illness–associated metabolic bone disease
      iv. Excessive phosphorus intake
      v. Vitamin D toxicity
      vi. Tumor lysis syndrome
      vii. Hypoparathyroidism
   d. Treatment:
      i. Reduce phosphorus intake (omit from PN solution, with or without reduction of ILE content of PN solution if a high-fat formulation [controversial as phosphorus in organic form: phospholipids and not inorganic such as sodium phosphate]; change to "low- or no-electrolyte" renal enteral formula).
      ii. Phosphate binders if consuming oral/enteral nutrition (EN) (Am J Health Syst Pharm 2005;62:2355-61). Dosing should be timed with meals for patients consuming oral diets or given at regular intervals before and/or during enteral nutrition administration for patients on EN.

Table 17. Phosphate Binders

Drug	Available Strength	Phosphorus-Binding Capacity	Recommended Empiric Initial Dose
Calcium carbonate[a]	500, 750, 1000 mg per tablet	Calcium 43 mg/g	1 g QID
Calcium acetate[a]	667 mg per tablet	Calcium 106 mg/g	1334–2001 mg TID
Sevelamer[b]	400 mg, 800 mg per tablet; 800 mg or 2400 mg per packet (for oral suspension)	Sevelamer 80 mg/g	800–1600 mg TID
Lanthanum[c]	500, 750, 1000 mg per tablet or packet	Lanthanum 78–156 mg/g	500 mg TID

[a]Do not use if the patient is hypercalcemic.

[b]Sevelamer carbonate comes in a powder for suspension; easier for administering to patients receiving EN; less likely to worsen metabolic acidosis than tablet in patients with renal failure because tablet is in hydrochloric acid salt form.

[c]750 and 1000 mg available as oral packet; 500, 750, and 1000 mg available as chewable tablet.

EN = oral/enteral nutrition; QID = four times daily; TID = three times daily.

II. ACID-BASE DISORDERS

A. Normal Homeostasis
   1. Normal values

Table 18. Normal Blood Gas Values

Variables	Arterial Blood	Mixed Venous Blood
pH	7.35–7.45	7.31–7.41
P_{CO_2}	35–45	41–51
P_{O_2}	80–100	35–40
HCO_3	22–26	22–26
Base excess	-2 to +2	-2 to +2
O_2 saturation	> 95%	70%–75%

HCO_3 = bicarbonate.

   2. Interpreting ABGs
      a. Acidemia (pH less than 7.35) versus alkalemia (pH greater than 7.45)
      b. Acidemia and alkalemia refer to an abnormal pH being either low or high, respectively. Acidosis and alkalosis refer to the metabolic or respiratory processes that led to the abnormal pH. Although the terms *emia* and *osis* are similar, they are different.
      c. For simple acid-base disorders, identify pH, P_{CO_2}, and HCO_3 in that order. Whichever side of 7.40 the pH is on, the respiratory or metabolic processes that coincide with that pH abnormality are the primary etiology. If the pH is less than 7.40, an elevated P_{CO_2} (respiratory acidosis) or a decreased HCO_3 (metabolic acidosis) is the primary etiology. If the pH is greater than 7.40, a decreased P_{CO_2} (respiratory alkalosis) or an increased HCO_3 (metabolic alkalosis) is the primary etiology. An easy introductory overview to acid-base disorders by Haber is provided in the references (West J Med 1991;155:146-51).
      d. Sometimes more than one primary abnormality is present, or the anticipated compensatory process (metabolic or respiratory) is inadequate and may be contributing to the acid-base disorder. As a result, various formulas have been developed to predict what may be considered adequate compensation. However, these mathematical equations have limitations in their clinical utility and accuracy (J Trauma Acute Care Surg 2012;73:27-32; Clin J Am Soc Nephrol 2007;2:162-74; Crit Care Med 2007;35:1264-70; Am J Respir Crit Care Med 2000;162:2246-51; Arch Intern Med 1992;152:1625-9; J Crit Care 2013;28:1103; N Engl J Med 2014;371:1434-45) and can be difficult to memorize (West J Med 1991;155:146-51).
      e. In addition, the issue of mixed acid-base disorders is confounded by several factors that can lead to errors in interpreting acid-base disorders. These include non–steady-state conditions as well as the inability for the patient to adequately compensate through the respiratory pathway because of mechanical ventilator restrictions. Some of the more common equations for assessing acid-base disorders are discussed later in this chapter.

Box 1. Adverse Effects of Severe Acidemia (pH 7.25 or less)

Impaired cardiac output
Increased metabolic demands
Peripheral ischemia (centralization of blood)
Insulin resistance and hyperglycemia
Increased pulmonary vascular resistance
Decreased ATP synthesis
Hypotension
Increased protein breakdown
Increased risk of arrhythmias
Hyperkalemia
Decreased catecholamine responsiveness
Diaphragmatic fatigue/dyspnea
Obtundation/coma
Hyperventilation

Box 2. Adverse Effects of Severe Alkalemia (pH of 7.55 or greater)

Arteriolar constriction
Hypokalemia
Hypotension
Hypophosphatemia
Increased risk of arrhythmias
Hypocalcemia/tetany
Decreased coronary blood flow/decreased angina threshold
Organic acid production
Seizures
Hypoventilation/hypercapnia/hypoxemia
Lethargy, delirium, stupor

3. Use of base excess
    a. Reflects the amount of base needed in vitro to return the plasma pH to 7.40 at standard conditions (P_{CO_2} 40 mm Hg, 37°C)
    b. Base excess reflects the metabolic component to interpreting the ABG. Despite its simplicity, it has limitations (J Trauma Acute Care Surg 2012;73:27-32). Crystalloid resuscitation (leading to hyperchloremic acidosis), exogenous bicarbonate (HCO_3) administration, ethanol ingestion, and acetate/HCO_3 buffer in hemodialysis or CRRT solutions can lead to erroneous base excess calculations and errors in interpretation (J Trauma Acute Care Surg 2012;73:27-32).
4. Compensatory response to acid-base disorders

Table 19. Anticipated Compensation to Acid-Base Disorders

Primary Disorder		Serum Bicarbonate (mEq/L)	Anticipated P_{CO_2} (mm Hg)
Metabolic acidosis		< 22	$(1.5 \times HCO_3) + 8 (\pm 2)$
Metabolic alkalosis		> 26	$(0.7 \times HCO_3) + 20$
Primary Disorder		**P_{CO_2} (mm Hg)**	**Anticipated Serum Bicarbonate (mEq/L)**
Respiratory acidosis	Acute[a]	> 42	HCO_3 should increase by 1 mEq/L for every 10 mm Hg increase in P_{CO_2} above 40 mm Hg
	Chronic[a]		HCO_3 should increase by 4–5 mEq/L for every 10 mm Hg increase in P_{CO_2} above 40 mm Hg
Respiratory alkalosis	Acute[a]	< 38	HCO_3 should decrease by 2 mEq/L for every 10 mm Hg decrease in P_{CO_2} below 40 mm Hg
	Chronic[a]		HCO_3 should decrease by 4–5 mEq/L for every 10 mm Hg decrease in P_{CO_2} below 40 mm Hg

[a]Compensation is different for acute versus chronic respiratory disorders because it takes about 2 days for the kidneys to adapt to a persistent change in respiratory status.

Information from: Berend K, de Vries APJ, Gans ROB. Physiological approach of acid-base disturbances. N Engl J Med 2014;371:1434-45.

B. Respiratory Acidosis
   1. Common causes include pulmonary edema, pulmonary embolism, pneumonia, CNS depression, cardiac arrest, stroke, spinal cord injury, excessive sedation/analgesia, and overfeeding with PN/EN. Make sure it is not caused by excessive sedation/analgesia or overfeeding with EN/PN.
   2. Metabolic compensation – See Table 19.

C. Respiratory Alkalosis
   1. Common causes include uncontrolled pain, nicotine and drug withdrawal, agitation, pneumonia, stimulant drugs, salicylate toxicity (due to direct respiratory stimulation), and head injury. Make sure the patient is getting adequate sedation/analgesia, fever/pneumonia is being treated; nicotine and drug withdrawal regimen is/are appropriate.
   2. Metabolic compensation – See Table 19.

D. Metabolic Acidosis
   1. Use of the serum anion gap (AG)
      a. Used to determine the etiology for the metabolic acidosis. AG is the difference between major cations and anions in blood (trying to detect whether there is an abundance of unmeasured anions). If an AG is present, then a metabolic acidosis is present, regardless of pH or HCO_3.

$$AG = Na - (Cl + HCO_3)$$

      b. Normal range is about 3–14 mEq/L. Some clinicians will include serum potassium when calculating cations (and the normal AG will need to be adjusted), but this is uncommon.
      c. Adjust AG for serum albumin (Crit Care Med 1998;26:1807-10). The difference in serum albumin concentration (grams per deciliter) from normal should be multiplied by 2–2.5 and added to the anions (chloride and bicarbonate).

$$\text{Albumin adjusted AG} = Na - Cl - HCO_3 - (2.5 \times [4 - \text{serum albumin}]).$$

      Some clinicians also adjust for serum phosphorus (Crit Care Med 2007;35:2630-6). Serum phosphorus (milligrams per deciliter) can be multiplied by 0.5 and added to anions, and lactate can also be included but is not common in routine clinical practice. Using this method (and including

serum potassium), the adjusted AG (or sometimes called the strong ion gap when referring to the physicochemical methodology for interpreting acid-base disorders) should be close to 0 (± 2) if the patient does not have an AG acidosis.

d. Causes of an AG acidosis: One easy pneumonic to remember (there are others) is A MUD PIE:
A = Aspirin (or other salicylates)
M = Methanol
U = Uremia (including rhabdomyolysis)
D = Diabetes (diabetic ketoacidosis [DKA]*) *Incidence of euglycemic DKA has grown with increase in use of sodium-glucose transporter 2 inhibitors.
P = Paraldehyde, propylene glycol
I = Infection or ischemia (lactic acidosis)
E = Ethylene glycol or ethanol toxicity

e. Types of lactic acidosis (lactate greater than 4 mmol/L and pH less than 7.35)
   i. Type A: Hypoperfusion (cardiogenic or septic shock, regional ischemia, severe anemia)
   ii. Type B: Metabolic – No tissue hypoxia
      (a) B1 = underlying disease (diabetes, liver disease, leukemia, lymphoma, AIDS)
      (b) B2 = drugs/toxins (metformin, didanosine/stavudine/zidovudine, ethanol, linezolid, propofol, propylene glycol toxicity caused by intravenous lorazepam or pentobarbital), nitroprusside (cyanide) toxicity
      (c) B3 = inborn errors of metabolism (pyruvate dehydrogenase deficiency)

f. Causes of a normal AG acidosis
Another easy pneumonic to remember (there are others) is ACCRUED.
A = Ammonium chloride/acetazolamide (urine bicarbonate loss)
C = Chloride intake (PN, intravenous solutions)
C = Cholestyramine (GI bicarbonate loss)
R = Renal tubular acidosis: Types I, II, and IV
U = Urine diverted into the intestine (e.g., ileal conduit, vesicoenteric fistula)
E = Endocrine disorders (e.g., aldosterone deficiency)
D = Diarrhea or small/large bowel fluid losses (e.g., enterocutaneous fistulas)

2. In the presence of an elevated AG, the delta ratio can be assessed for determining mixed acid-base disorders
Delta ratio =
$\Delta AG/\Delta HCO_3$ = (measured AG − normal AG)/(normal HCO_3 − measured HCO_3) =
(AG−14)/(24 − measured HCO_3)

Table 20. Interpreting Delta Ratio[a]

Delta Ratio	Assessment
< 0.4	Hyperchloremic normal AG acidosis
< 1	Mixed high AG acidosis and hyperchloremic normal AG acidosis
1–2	AG acidosis (no hidden process)
> 2	High AG acidosis and concurrent metabolic alkalosis OR a preexisting compensated respiratory alkalosis

[a]The ratio should be used cautiously in interpreting mixed acid-base disorders, given that it is associated with poor sensitivity because of several influencing factors (J Am Soc Nephrol 2007;18:2429-31). This author prefers to look at current clinical state/diagnoses and recent therapeutic interventions for the patient to ascertain whether a mixed acid-base disorder is potentially present.

3. An alternative method (and perhaps a simpler approach) to the delta ratio is to calculate the "excess gap" compared with the AG (West J Med 1991;155:146-51).
   Excess gap = AG – 14.
   a. The excess gap is then added to the measured serum bicarbonate concentration.
   b. If the sum is less than a normal serum bicarbonate concentration (e.g., 28–30 mEq/L), a mixed AG and non-AG acidosis is present.
   c. If the sum is greater than a normal HCO_3 concentration, the patient likely has an AG acidosis and concurrent metabolic alkalosis.
4. Evaluation of respiratory compensation: See Table 19.
5. Treatment
   a. Aggressive interventional therapy unnecessary until pH less than 7.20–7.25
   b. Treat primary etiology! This should be the focus of treating the acid-base disorder.
   c. Intravenous sources of alkali – Done conservatively in conjunction with treating primary disorder whenever possible. The intent is not to normalize the pH but to improve the pH (definitely avoid overcorrection).
      i. Sodium bicarbonate – Most commonly used
      ii. Sodium acetate – Available in PN solutions and compounded intravenous fluids
      iii. Sodium citrate – Used orally for patients with chronic kidney injury
   d. Total bicarbonate dose (mEq) = 0.5 x Wt (kg) x (24 - HCO_3)
      i. Give one-third to one-half of the calculated total dose (or 1–2 mEq/kg) for several hours to achieve a pH of around 7.25 (avoid boluses if possible).
      ii. Once the pH is around 7.25 or greater, slower correction without increasing bicarbonate more than 4–6 mEq/L to avoid exceeding the target pH
      iii. Serial ABGs (e.g., every 6 hours); watch rate of decrease in serum potassium and calcium
      iv. Use of sodium bicarbonate injection is controversial in patients with lactic acidosis (Curr Opin Crit Care 2008;14:379-83).
      v. The BICAR-ICU study (Lancet 2018;392:31-40) evaluated 389 critically ill patients with metabolic acidemia in a multicenter, intention-to-treat trial in which patients were randomized to sodium bicarbonate therapy or placebo. Most patients had an elevated serum lactate at enrollment. The primary outcome (composite of death by day 28 and presence of at least one organ failure at day 7) was not statistically significant. However, in the prespecified stratum of patients with AKI, the primary outcome was decreased in the treatment group (37% vs. 54%, p=0.0283).
   e. Adverse effects of excess sodium bicarbonate:
      i. Hypernatremia, hyperosmolality, volume overload
      ii. Hypokalemia, hypocalcemia, hypophosphatemia
      iii. Paradoxical worsening of the acidosis (if the fractional increase in P_{CO_2} production exceeds the fractional bicarbonate change)
      iv. Over-alkalinization

Patient Case

Questions 8 and 9 pertain to the following case.
A 60-year-old woman (weight 80 kg) was admitted to the hospital after 1 week of severe diarrhea. She presents with clinical evidence of dehydration (hypotension, tachycardia, decreased urine output) and is weak. Her serum laboratory values are as follows: Na 145 mEq/L, K 3.0 mEq/L, Cl 118 mEq/L, total CO_2 18 mEq/L, BUN 29 mg/dL, SCr 0.9 mg/dL, glucose 122 mg/dL, calcium 9.1 mg/dL, phosphorus 3.7 mg/dL, magnesium 1.4 mg/L, albumin 3.9 g/dL, and lactate 1.6 mmol/L. Her ABG is pH 7.29, P_{O_2} 93 mm Hg, P_{CO_2} 34 mm Hg, HCO_3 17 mEq/L, and base excess -5 mEq/L. She has a 30 pack/year smoking history.

8. Which best describes the patient's type of acid-base disorder?
   A. Hyperchloremic, normal AG acidosis
   B. AG acidosis
   C. AG acidosis with hyperchloremia
   D. Respiratory alkalosis with concurrent metabolic alkalosis

9. Which is the most appropriate initial fluid therapy for this patient?
   A. 0.45% sodium chloride with potassium chloride 20 mEq/L
   B. 0.9% sodium chloride with potassium chloride 20 mEq/L
   C. Lactated Ringer solution
   D. 5% dextrose

E. Metabolic Alkalosis: pH greater than 7.45; symptoms are not usually severe until pH is greater than 7.55–7.60.
   1. Assessment (to help guide treatment) based on urinary chloride
      a. Saline responsive (urinary chloride less than 10 mEq/L)
         i. Excessive gastric fluid losses
         ii. Diuretic therapy (especially loop diuretics)
         iii. Dehydration (contraction alkalosis)
         iv. Hypokalemia
         v. (Over-) Correction of chronic hypercapnia
      b. Saline resistant (urinary chloride greater than 20 mEq/L)
         i. Excessive mineralocorticoid activity (e.g., hydrocortisone, fludrocortisone)
         ii. Excessive alkali intake
         iii. Profound potassium depletion (serum potassium less than 3 mEq/L)
         iv. Excess licorice (glycyrrhizic acid, a mineralcorticoid) intake
         v. Massive blood transfusion
      c. Respiratory compensation (highly variable and may not be possible for ventilator-dependent patients)
      d. Intravascular volume status (important for saline-responsive alkalemia)
   2. Treatment – Saline-responsive alkalemia
      a. Treat underlying cause (if possible).
      b. Decreased intravascular volume: Give intravenous 0.9% sodium chloride, Plasmalyte/Normosol, or Lactated Ringer's infusion (with potassium chloride, if necessary).
      c. Increased intravascular volume: Acetazolamide 250–500 mg orally or intravenously once to four times daily plus potassium chloride if necessary (Intensive Care Med 2010;36:859-63; Crit Care Med 1999;27:1257-61; Acta Anaesthesiol Scand 1983;27:252-4).
         i. Hydrochloric acid therapy if alkalosis is refractory or initial pH greater than 7.6

(a) 0.1 N or 0.2 N of hydrochloric acid (use 0.2 N for patients requiring fluid restriction). Hydrochloric acid should be given by central venous administration, and the solution must be in a glass bottle.

ii. Dosage of hydrochloric acid:

(a) Chloride deficit (Arch Surg 1975;110:819-21):
Dose (mEq) = 0.2 L/kg x Wt (kg) x (103 − serum chloride)

(b) Bicarbonate excess (J Am Soc Nephrol 2000;11:369-75):
Dose (mEq) = 0.5 L/kg x Wt (kg) x (serum bicarbonate -24)

(c) A practical approach is to administer one-half of the calculated dose over 12 hours and repeat the arterial blood gas at 6 and 12 hours with readjustment of infusion rate, if necessary. Discontinue the infusion when the pH is less than 7.5.

3. Treatment – Saline-unresponsive alkalosis: Treat underlying cause (if possible).
   a. Exogenous corticosteroids – Decrease dose or use drug with less mineralocorticoid effect.
   b. Excessive alkali intake – Alter regimen.
   c. Profound hypokalemia (serum potassium less than 3 mEq/L) – Aggressive potassium supplementation
   d. Rare causes: Endogenous mineralocorticoid excess (Bartter or Gitelman syndrome) – Spironolactone, amiloride, or triamterene; consider surgery
   e. Liddle syndrome: Amiloride or triamterene

III. NUTRITION SUPPORT

A. Nutritional Assessment
   1. Nutrition Risk Assessment
      a. "Determination of Nutrition Risk" – recommended by 2016 SCCM/ASPEN guidelines to guide the approach to nutrition support therapy in critically ill patients. High nutrition risk identifies those patients most likely to benefit from early EN therapy. (JPEN J Parenter Enteral Nutr 2016;40: 159-211)
         i. Nutritional Risk Screening (NRS 2002; Clin Nutr 2003;22:321-36)

Table 21. Determination of Nutrition Risk by NRS 2002 score

Nutritional Status		Severity of Disease	
	Score		**Score**
Normal Nutritional Status	0	Normal nutritional requirements	0
Wt loss > 5% in 3 months or food intake < 50% to 70% of normal in preceding week	1	Hip fracture, chronic patients with acute complications (e.g., cirrhosis, COPD, diabetes, oncology)	1
Wt loss > 5% in 2 months or body mass index (BMI) 18.5 to 20.5 + impaired general condition or food intake < 25% to 50% of normal in preceding week	2	Major abdominal surgery, stroke, pneumonia, hematologic malignancy	2
Wt loss > 5% in 1 month (~15% in 3 months) or BMI < 18.5 + impaired general condition or food intake 0 to 25% of normal in preceding week	3	Head injury, bone marrow transplantation, critically ill ICU patient	3
Add the two scores (nutritional status + severity of disease). If age 70 years or greater, add 1 to total score. If age-corrected score 3 or greater, start nutritional support.			

ii. Nutrition Risk in the Critically Ill (NUTRIC) score (Crit Care 2011;15:R268)

Table 22. Determination of Nutrition Risk by the NUTRIC Score

Variable	Range	Points
Age, yrs	< 50 50 - 74 > 74	0 1 2
APACHE II	< 15 15 -19 20 - 28 > 28	0 1 2 3
SOFA	< 6 6 – 9 > 9	0 1 2
Number of comorbidities	0 - 1 > 1	0 1
Days from hospital to ICU admission	0 - < 1 ≥ 1	0 1
IL-6	0 - < 400 ≥ 400	0 1
High Score = 6 to 10 points: Associated with worse clinical outcomes and are most likely to benefit from aggressive nutrition therapy Low Score = 0 to 5 points: These patients have a low nutrition risk		

iii. Modified NUTRIC Score (Clin Nutr 2016;35:158-62). Because IL-6 is not routinely available for assessment, a modified NUTRIC (mNUTRIC) score has been validated, which omits IL-6. An mNUTRIC score of 5 or greater is considered high nutrition risk.

2. Malnutrition categories
   a. American Society for Parenteral and Enteral Nutrition (ASPEN) International Consensus Nomenclature (JPEN J Parenter Enteral Nutr 2010;34:156-9)
      i. Starvation-related malnutrition (e.g., anorexia nervosa)
      ii. Chronic disease-related malnutrition (e.g., inflammatory bowel disease, organ failure)
      iii. Acute disease or injury-related malnutrition (e.g., major infection, burns, trauma)
   b. "Classic" definition – Not in favor with conventional assessment techniques and tools
      i. Marasmus (e.g., decreased fat/muscle protein stores but normal serum proteins)
      ii. Kwashiorkor (e.g., normal fat, decreased muscle protein, and serum proteins)
      iii. Kwashiorkor-Marasmus mix (decreased fat, muscle protein, and serum proteins)
   c. Based on body mass index (BMI) = weight (kg)/height^2 (m^2) – Should not be used alone to identify malnutrition
      i. Less than 18.5 kg/m^2: Underweight
      ii. 18.5–24.9 kg/m^2: Normal
      iii. 25–29.9 kg/m^2: Overweight
      iv. 30–34.9 kg/m^2: Class I obesity
      v. 35–39.9 kg/m^2: Class II obesity
      vi. Greater than 40 kg/m^2: Class III obesity
   d. Academy of Nutrition and Dietetics/American Society for Parenteral and Enteral Nutrition Consensus Malnutrition Characteristics (JPEN J Parenter Enteral Nutr 2012;36:275-83)
      i. Diagnosed by the presence of two or more of the following:
         (a) Insufficient energy intake

(b) Weight loss
(c) Loss of muscle mass
(d) Loss of subcutaneous fat
(e) Localized or generalized fluid accumulation (may sometimes mask weight loss)
(f) Diminished functional status as measured by hand grip strength
(1) In the context of these criteria, hand grip strength intended to be used to document decline in physical function as appropriate to patient circumstances

ii. Nonsevere (moderate) or severe malnutrition determined by duration and severity of above symptoms, which is further categorized as:
(a) Acute illness or injury-related malnutrition
(b) Chronic disease-related malnutrition
(c) Social or environmental-related malnutrition

iii. Visceral protein concentrations (e.g., albumin, prealbumin, transferrin) are not reliable indicators and not included in this definition. They should not be used to assess nutrition status in critically ill patients (JPEN J Parenteral Enteral Nutr 2016;40:159-211; Nutr Clin Pract 2021;36:22-8).

iv. BMI is not included in this classification because patients become malnourished regardless of body habitus.

3. Empiric weight adjustment for amputations (Table 23)

Table 23. Body Compartments' Contribution to Body Weight (J Am Diet Assoc 1995;95:215-8)

Body Part Amputation	Approximate Contribution to Body Weight (%)
Foot	1.5
Calf, foot	5.9
Leg (from hip)	16
Hand	0.7
Hand and forearm	2.3
Arm	5

B. Energy Requirements

Assessing caloric requirements: Indirect calorimetry – Measured energy expenditure by oxygen consumption and CO_2 production – The "gold standard"

1. Respiratory quotient (V_{CO_2}/V_{O_2}); 1 for carbohydrate oxidation; 0.7 for fat oxidation; 0.8 for protein oxidation; greater than 1 usually implies overfeeding (net fat synthesis), less than 0.7 suggests ketosis or an error in measurement (too much fraction of inspired oxygen [F_{IO_2}] variability at higher F_{IO_2} concentrations). Widespread use in clinical practice is limited by availability and cost of indirect calorimeters. Furthermore, accuracy of measurements is affected by many common factors in the ICU (presence of chest tubes, use of supplemental oxygen, FiO_2 settings, PEEP settings, continuous renal replacement therapy, anesthesia, movement).
2. Organization guideline recommendations

Table 24. Guideline Recommendations for Caloric Intake

Organization	Recommendation
American College of Chest Physicians (Chest 1997;111:769-78)	25 kcal/kg/day; increase by 10%–20% with SIRS
SCCM/ASPEN (JPEN J Parenter Enteral Nutr 2016;40:159-211)	25–30 kcal/kg/day or use of indirect calorimetry (lower amounts may be given to medical ICU patients with ARDS for the first week) Obesity: Goal EN regimen should not be > 65%–70% target energy requirements as measured by indirect calorimetry. For BMI 30–50 kg/m^2, a weight-based equation of 11–14 kcal/kg actual body weight/day is recommended. For BMI > 50 kg/m2, 22–25 kcal/kg ideal body weight/day is recommended
ASPEN (JPEN J Parenter Enteral Nutr 2022;46:12-41)	In the first 7–10 days of ICU stay, 12–25 kcal/kg/day (recommended because of no findings of significant difference in clinical outcomes between patients with higher vs. lower intake; however, this does not take into account nutrition risk, presence of malnutrition, ICU patient population [most supporting literature is in medical ICU], or that most patients on EN will receive substantially less than goal intake during this time because of frequent interruptions to feeding for ICU care)
ESPEN (Clin Nutr 2023;42:1671-89)	20–25 kcal/kg/day (in the absence of measured REE)
ESPEN PN (Clin Nutr 2009;28:387-400)	25 kcal/kg/day (in the absence of measured REE)
ESPEN EN (Clin Nutr 2006;25:210-23)	20–25 kcal/kg/day (during acute phase) 25–30 kcal/kg/day (during recovery)
ASPEN Obesity Guidelines (JPEN J Parenter Enteral Nutr 2013;37:714-44)	< 14 kcal/kg actual weight/day or 50%–70% of estimated requirements when given with a high-protein intake
Eastern Association for Surgery of Trauma (J Trauma 2004;57:660-9)	25–30 kcal/kg/day or 1.2–1.4 x BEE (Harris-Benedict equations) 30 kcal/kg/day (1.4 x BEE) for patients with TBI 22–25 kcal/kg/day for patients with paraplegia 20–22 kcal/kg for patients with quadriplegia

BEE = basal energy expenditure; ESPEN = European Society for Clinical Nutrition and Metabolism; REE = resting energy expenditure; SCCM = Society of Critical Care Medicine; SIRS = systemic inflammatory response syndrome.

3. Predictive methods
    a. Mifflin-St. Jeor equations (preferred for non–ventilator-dependent patients with obesity, AKI or CKD, or hepatic encephalopathy when a hypocaloric, high-protein regimen is not possible)
        i. Women = (10 x Wt) + (6.25 x Ht) – (5 x Age) − 161*
        ii. Men = (10 x Wt) + (6.25 x Ht) − (5 x Age) + 5*
        *Age (years); Ht (centimeters); Wt = actual body weight (kilograms).
    b. Penn State equation (preferred for ventilator-dependent patients with obesity):
    REE = (Mifflin x 0.96) + (Tmax x 167) + (Ve x 31) − 6212*
    *REE = resting energy expenditure; Tmax = maximum temperature in degrees Celsius; Ve = minute ventilation, liters per minute.
    c. Modified Penn State equation (preferred for ventilator-dependent patients with obesity 60 years or older
    REE = (Mifflin x 0.71) + (Tmax x 85) + (Ve x 64) − 3085*

d. Basal energy expenditure (BEE) – Harris-Benedict equations (preferred for small adults and older adult patients)
   i. Women = 655 + (9.6 x Wt) + (1.7 x Ht) − (4.7 x Age)*
   ii. Men = 66 + (13.7 x Wt) + (5 x Ht) − (6.8 x Age)*
   *Wt (kilograms), Ht (centimeters), Age (years).

4. Adverse effects of overfeeding – Do not exceed 4–5 mg/kg/minute of glucose/carbohydrate in the acute phase of critical illness (Ann Surg 1979;190:274-85); limit total caloric intake (do not exceed 1.3–1.5 x measured resting energy expenditure or 25–30 kcal/kg/day for most patients); do not exceed soybean oil (SO)-ILE intake of 2.5 g/kg/day; most clinicians limit SO-ILE in critically ill patients to around 1 g/kg/day or less.
   a. Hypercapnia: It was traditionally thought that excessive glucose intake alone was responsible for hypercapnia observed during overfeeding. However, studies of acutely ill patients showed that aggressive feeding resulted in marked increases in CO_2 production (Ann Surg 1980;191:40-6; JAMA 1980;243:1444-7). Substitution of glucose kilocalories with lipid decreases CO_2 production (Anesthesiology 1981;54:373-7) when overfeeding but does not alter CO_2 production if not overfeeding (e.g., 1.3 x BEE) (Chest 1992;102:551-5). Because most institutions lack the ability to measure energy expenditure, estimates are used. If the patient experiences hypercapnia without a known cause, the nutrition therapy should be suspected and the caloric intake empirically decreased (especially if the patient is having difficulty weaning from the ventilator).
   b. Hyperglycemia: In a retrospective study of 102 patients receiving PN and who were not predisposed to hyperglycemia, dextrose intakes in excess of 5 mg/kg/minute resulted in substantial hyperglycemia (blood glucose [BG] greater than 200 mg/dL) in 18 of 37 patients (Nutr Clin Pract 1996;11:151-6). Patients with stress-induced hyperglycemia or diabetes are even more susceptible to hyperglycemia with EN or PN. This is one reason (of many) that critically ill patients who require PN should receive PN via continuous infusion rather than cyclic infusion (i.e., over 12–14 hour/night), even if they tolerated a cyclic infusion of PN prior to ICU admission (Nutr Clin Pract 2023;38:1263-72).
   c. Fatty infiltration of the liver: May be because of overfeeding with fat or carbohydrate. Usually presents as a cholestatic liver disease (increased gamma-glutamyl transferase (GGT), alkaline phosphatase, and ultimately bilirubin) after at least 1 week to 10 days of overfeeding (Arch Surg 1978;113:504-8). May be transient or reversible or can progress to end-stage liver disease. Throughout a few weeks, patients can appear jaundiced. Patients with critical illness and/or infections tend to be more susceptible to hepatic steatosis compared with non–critically ill patients (possibly because of an exaggerated inflammatory process). Although treatment with fish oil appears promising in infants and children (Nutr Clin Pract 2013;28:30-9; Ann Surg 2009;250:395-402), data for adults are limited. Usual treatment for adult patients with suspected PN-associated liver disease is first to ensure that the patient is not being overfed. Although evidence is lacking, most nutrition support clinicians would agree that if 100% SO-ILE is being used, consideration of an ILE formulation with a lower percentage of SO is warranted (refer PN formulation section for description of ILE formulations). Cyclic PN (when PN is infused over fewer hours per day) may be considered for patients who are no longer critically ill. However, cyclic PN is typically not appropriate for critically ill patients due to the high glucose infusion rate and large volumes of fluid (relative to time) required for administration. Reinstituting EN as soon as possible (if possible) is of utmost importance.

C. Protein Requirements
   1. Guideline recommendations

Table 25. Guideline Recommendations for Protein Intake

Organization	Recommendation
American College of Chest Physicians (Chest 1997;111:769-78)	1.2–1.5 g/kg/day 1.5–2 g/kg/day not to exceed 2 g/kg/day with SIRS Routine nitrogen balance (NB) determinations recommended
SCCM/ASPEN (JPEN J Parenter Enteral Nutr 2016;40:159-211)	1.2–2 g/kg/day; higher amounts are likely needed for multiple trauma or burns; higher amounts are needed during continuous renal replacement therapy (CRRT), up to 2.5 g/kg/day For patients with open abdomen, provide an additional 15–30 g of protein per liter of output from wound drain(s) or negative-pressure wound devices (caused by exposure of peritoneum, which produces high-protein exudate) For patients with obesity receiving a hypocaloric (low energy, high protein) regimen: 2 g/kg/day IBW for BMI 30–40 kg/m^2, up to 2.5 g/kg/day IBW for BMI ≥ 40 kg/m^2
ESPEN ICU (Clin Nutr 2023;42:1671-89)	1.3 g/kd/day
ESPEN PN (Clin Nutr 2009;28:387-400)	1.3–1.5 g/kg IBW/day
ESPEN EN (Clin Nutr 2006;25:210-23)	Not given
ASPEN Obesity Guidelines (JPEN J Parenter Enteral Nutr 2013;37:714-44)	1.2 g/kg actual weight/day or 2–2.5 g/kg IBW/day when given with a hypocaloric regimen
Eastern Association for Surgery of Trauma (J Trauma 2004;57:660-9)	1.25–2 g/kg/day; 2 g/kg/day for burns
International Protein Summit Consensus recommendations (Nutr Clin Pract 2017;32(suppl 1):142S-151S) – in reference to ICU patients	Minimum of 1.2 g/kg/day with doses up to 2–2.5 g/kg/day Obese (BMI 30–39.9 kg/m2): 2 g/kg IBW/day Obese (BMI ≥ 40 kg/m2): 2.5 g/kg IBW/day AKI: 1.2–2 g/kg/day CRRT: 1.5–2.5 g/kg/day Age > 60 yr: 2–2.5 g/kg/day Persistent inflammation catabolism syndrome: 1.2–2 g/kg/day Infants and children: A minimum of 1.5 g/kg/d

2. Does more protein really make a difference?
    a. Weijs et al. (2012) (JPEN J Parenter Enteral Nutr 2012;36:60-8): 28-day mortality improved in those who received an average of 1.3 g/kg/day versus 1.1 or 0.8 g/kg/day (886 mixed ICU patients).
    b. Allingstrup et al. (2012) (Clin Nutr 2012;31:462-8): 28-day mortality improved in those who received an average of 1.5 g/kg/day versus 1.1 or 0.8 g/kg/day (113 mixed ICU patients).
    c. JPEN J Parenter Enteral Nutr 2016;40:45-51: Improved survival with protein intake greater than 80% of goal (1.2 g/kg/day) in 2828 mixed ICU patients. Achieving 80% of caloric goals did not affect mortality.
    d. Crit Care Med 2017;45:156-63: Mortality was decreased with increased protein intake for high-risk (NUTRIC score greater than 4) mixed ICU patients with a prolonged ICU stay of more than 11 days. Low-risk patients were unaffected.

e. ASPEN guidelines for nutrition support in adult critically ill patients (2022) (JPEN J Parenter Enteral Nutr 2022;46:12-41): Based on the low number of high-quality trials available for evaluation, no difference in clinical outcomes has been observed with provision of higher versus lower protein intake.
f. EFFORT Protein Trial (2023) (Lancet 2023;401:568-576) No difference in incidence of alive at hospital discharge between patients who received an average protein intake of 1.6 versus 0.9 g/kg/day (high protein versus usual dose protein, respectively). Subgroup analyses on alive at discharge and 60-day mortality favored usual dose protein group for patients with AKI (1301 mixed ICU patients, 83.6% study population in medical ICU).

D. Principles of EN and PN
1. Indications for EN: If the patient is unable to eat adequate amounts to achieve goal nutritional intake. EN is preferred to PN because EN helps maintain the integrity of the GI tract, has fewer infectious complications, is associated with decreased ICU length of stay, and is more cost-effective (JPEN J Parenter Enteral Nutr 2009;33:277-316; Ann Surg 1992;215:503-13; JPEN J Parenter Enteral Nutr 2016;40:159-211). This position has been challenged recently because of newer literature (JAMA 2013;309:2130-8; Lancet 2013;381:385-93; N Engl J Med 2014;371:1673-84; Crit Care 2016;20:117; Clin Nutr 2017;36:623-50) indicating a reduction in infectious complications with PN as compared with the past literature (likely because of improvements in catheter care, PN formula management and compounding practices, and glycemic control standards over the past several decades). In fact, the 2022 ASPEN guidelines for critically ill patients recommends that either PN or EN is acceptable as the primary feeding modality in the first week of critical illness (evidence GRADE: high; strength of GRADE recommendation: strong) (JPEN J Parenter Enteral Nutr 2022;46:12-41). This is based on findings that similar energy intake provided as PN is not superior to EN and no differences in harm were shown. Patient-specific factors and clinical judgment should guide the choice between PN versus EN.
   a. Lack of bowel sounds, flatus, or bowel movement is not a contraindication for EN because these are nonspecific indicators of GI function (JPEN J Parenter Enteral Nutr 2009;33:277-316; SCCM/ASPEN 2009).Evidence of ileus (e.g., dilated loops of bowel on abdominal radiography) is, however, a relative contraindication for EN.
   b. High NG output (greater than around 800 mL NG output) in a 24-hour period may indicate delayed gastric emptying, and the patient may not be ready for EN when fed into the stomach and post-pyloric feeding is not possible. High NG output may also indicate an ongoing ileus or obstruction, in which case, PN may be indicated.
   c. Refusal to eat/anorexia is not an absolute contraindication for EN. Ensure appropriate dietary preferences, and add high-calorie/protein liquid supplements to meals and bedtime snack first.
2. EN formulas

Table 26. EN Formulas

Product Category	Indication	Macronutrients	Additional Comments	Examples
Standard tube feeding	Minimal stress	1–1.2 kcal/mL Protein 40–50 g/L	Polymeric, 300–500 mOsm/kg, fiber	Jevity, Nutren 1.0, Fibersource HN
Volume restricted	Congestive heart failure, fluid-restricted patients	2 kcal/mL Protein 60–80 g/L	Polymeric, 700–800 mOsm/kg, fiber	Nutren 2.0, Resource 2.0, TwoCal HN

Table 26. EN Formulas (*Continued*)

Product Category	Indication	Macronutrients	Additional Comments	Examples
Renal	AKI (predialysis), renal dysfunction with increased serum potassium, phosphorus, magnesium	2 kcal/mL Protein 30–40 g/L	High calorie, low protein, no or low electrolytes, 600 mOsm/kg, volume restricted, fiber	Renalcal, Suplena
Renal	Renal failure with hemodialysis	1.8–2 kcal/mL Protein 80–90 g/L	High calorie, modest electrolytes, volume restricted, 1000 mOsm/kg, fiber	Novasource Renal, Nepro
Increased protein needs	Critically ill patients (especially trauma, surgical, burns)	1 kcal/mL Protein 62 g/L	High-protein content, fiber, isotonic	Replete, Promote
Glucose intolerance	Hyperglycemia, diabetes	1.2 kcal/mL Protein 60 g/L	Low-carbohydrate content, fiber	Diabetisource AC, Glucerna 1.2
Immune enhancing	Critically ill surgical and trauma patients, perioperative GI cancer	1.5 kcal/mL Protein 90–94 g/L	Additional arginine, glutamine; fish oil	Impact Peptide 1.5, Pivot
Bariatric	Patients with obesity with good renal function	1 kcal/mL Protein 93 g/L	High protein, low calories	Peptamen Intense VHP
Elemental	Malabsorption, fat intolerance	1–1.5 kcal/mL Protein 50–68 g/L	Low fat/MCT, di/tri-peptides/free AA, no fiber, low residue	Vivonex RTF, Vital
Protein supplement	High protein needs	10–15 g of protein per serving is typical (products vary)	Protein liquid, gel, or powder	Pro-Stat, ProMod

AA = amino acids; MCT = medium-chain triglycerides.

3. Enteral nutrition – Medication interactions: Hold EN 1 hour before and after drug administration. *Increase EN rate to account for time off EN.
    a. Warfarin (Pharmacotherapy 2008;28:308-13)
    b. Phenytoin (Nutr Clin Pract 1996;11:28-31)
    c. Levothyroxine (J Endocrinol Invest 2014;37:583-7; Nutr Clin Pract 2010;25:646-52) - In one study, liquid levothyroxine preparation was shown to alleviate need to hold EN. In lieu of holding EN, increasing the levothyroxine dose by 25 mcg per day with weekly monitoring of thyroid function tests has also been described. Care should be taken to reduce the dose back to the patient's home dose when they are transitioned off of EN if this option is chosen.
    d. Itraconazole (Antimicrob Agents Chemother 1997;41:2714-8)
    e. Fluoroquinolones (J Antimicrob Chemother 1996;38:871-6)
    *Some clinicians have empirically increased the dosage of these drugs while giving continuous enteral feeding rather than holding the EN for 1 hour before and after drug administration. With the exception of levothyroxine, this author discourages this practice, especially for warfarin and phenytoin, because the doses necessary to overcome the effects of drug binding to the continuous

EN are potentially toxic when the EN is held or discontinued without a dose adjustment. Others have increased the ciprofloxacin dose to 750 mg twice daily during continuous EN to achieve therapeutic plasma concentrations well above the MIC (minimum inhibitory concentration) for a gram-negative urinary tract infection (J Antimicrob Chemother 1996;38:871-6).

4. Indications for PN
    a. European Society for Clinical Nutrition and Metabolism (ESPEN) ICU guidelines (2023) (Clin Nutr 2023;42:1671-89): When oral nutrition and EN are contraindicated, PN should be initiated within 3–7 days in critically ill patients. Early PN (within 48 hours) can be provided instead of no nutrition if contraindications to EN are present in patients who are severely malnourished or at high nutrition risk.
    b. SCCM/ASPEN (2016) (JPEN J Parenter Enteral Nutr 2016;40:159-211) It is recommended to initiate PN as soon as possible after intensive care unit (ICU) admission if patients have contraindications to EN and are severely malnourished or at "high nutrition risk" as indicated by the NRS 2002 or NUTRIC score. PN should be avoided in the acute phase of sepsis regardless of the degree of nutrition risk because studies show longer hospital and ICU stays, longer duration of organ support, higher incidence of infectious complications, and higher hospital mortality with early supplemental and/or exclusive PN (JPEN J Parenter Enteral Nutr 2016;40:159-211). See a recent review on the controversial role of supplemental PN in adult patients (Nutr Clin Pract 2018;33:359-69).
    c. Other possible indications for PN: Severe, intractable vomiting or diarrhea, obstruction, impaired absorption (e.g., short bowel syndrome, high ostomy output), high-output enterocutaneous fistula (more than 500 mL/day), ischemic bowel, bowel discontinuity (JPEN J Parenter Enteral Nutr 2017;41:324-77)
    d. Summary: The approach depends on several factors (e.g., if the patient is malnourished or well nourished before ICU admission, patient acuity). Early nutrition (defined as within 24–72 hours according to published studies) appears to be beneficial for those with prolonged ICU stays and a high level of catabolism, including trauma, TBI, and thermal injury and for some surgical subpopulations. Impact of early nutrition appears more variable with respect to clinical outcome for medical ICU patients and is likely related to a shorter duration in ICU stay and a lower level of catabolism for many patients. Recent literature supports the safe use of PN as a substitute for EN (when EN is contraindicated or when EN delivery is inadequate) with no difference in the incidence of infections.
5. PN formulations
    a. Peripheral versus central venous administration
        i. Osmolarity of PN admixtures for peripheral administration is limited to less than 900 mOsm/L.
        ii. Because of osmolality restrictions, peripheral PN solutions are "diluted," requiring large volumes (contraindicated for fluid-restricted patients and difficult for older patients) and typically do not allow for adequate protein and energy provision.
        iii. Phlebitis is common with peripheral PN and it is difficult to use beyond 2–3 days in most types of peripheral access.
        iv. A central PN formulation should be delivered into a line with the catheter tip ending in a large diameter vein, such as the distal superior vena cava, adjacent to the right atrium. Central PN formulations are generally preferred for critically ill patients who require PN.
    b. Safe practice recommendations for prescribing PN solutions – Should be prescribed in total amount per day (e.g., dextrose 200 g/day, "brand X" amino acids 150 g/day, "brand Y" lipids 50 g/day, fluid volume 1800 mL/day, sodium chloride 60 mEq/day, potassium acetate 80 mEq/day), NOT by concentrations (e.g., 20% dextrose, 8% amino acids) or by compounding techniques (e.g., 500 mL of 50% dextrose in water plus 500 mL of 10% amino acids). Because different brands of amino acids and lipid injectable emulsions (ILE) contain different ingredients and have different compatibility and stability profiles, the brand name of these ingredients should also be noted in the PN prescription and label (JPEN J Parenter Enteral Nutr 2014;8:296-333).

c. Glucose requirements
   i. Obligatory requirements for CNS, renal medulla, bone marrow, leukocytes, etc.: Around 130 g/day
   ii. Surgical wound healing requires about 80–150 g/day (based on atrioventricular differences and blood flow from a burned limb)
   iii. Caloric contribution of glucose: 3.4 kcal/g (as opposed to carbohydrate 4 kcal/g)
   iv. Mean glucose oxidation rate in critically ill patients is around 5 mg/kg/day (or about 25 kcal/kg/day as glucose). In general, most clinicians avoid exceeding this glucose intake in the acute phase of critical illness.

d. ILE requirements
   i. SO-ILE products have historically been the main source of ILE in the United States. ILE may be given separately from the PN admixture or as part of the PN solution. When given separately from the dextrose/amino acid formulation, the maximum allowable hang time according to the FDA is 12 hours. Previous recommendations were to consider withholding SO-ILE during the first week following initiation of PN in critically ill patients (or limiting to 100 g/week during this time) (JPEN J Parenter Enteral Nutr 2016;40:159-211). This is due to the high omega-6 fatty acid (more pro-inflammatory) content of SO-ILE and limited evidence suggesting worse outcomes for ICU patients who receive ILE early in their ICU course. However, this evidence is very limited with several notable design flaws that likely confounded study outcomes (J Trauma 1997;43:52-60). Several ILE products are now commercially available in the United States, but owing to the limitations of available evidence, the 2022 ASPEN guidelines for nutrition support in critically ill patients do not make a recommendation for one ILE product over another. SMOFlipid consists of 30% soybean oil, 30% medium-chain triglycerides, 25% olive oil (OO), and 15% fish oil (FO). It may be more advantageous than 100% SO-ILE because of a decreased propensity to cause elevated serum triglyceride concentrations and decreased exposure to SO without compromising caloric intake. Additionally, limited data analyses suggest decreased infections in ICU patients and decreased liver function tests with prolonged PN use. Daily administration is usually required to meet essential fatty acid requirements (met only by the SO component). A 100% FO-ILE product is also FDA-approved for use in pediatric patients in the United States, and an OO,SO-ILE product (80% OO, 20% SO) is also approved for use in adults. All clinical and practical considerations listed previously should be evaluated when choosing which ILE product to use for critically ill patients.
   ii. Caloric contribution of ILE 10% = 1.1 kcal/mL (11 kcal/g); 20% = 2 kcal/mL (10 kcal/g); 30% = 3 kcal/mL (10 kcal/g)
   iii. Dosage: About 100–150 g of soybean oil weekly (or 1–1.5 g/kg weekly) is enough to prevent essential fatty acid deficiency (EFAD). The FDA states a maximum upper limit of 2.5 g/kg/day in adults, though SO-ILE provision in critically ill patients should generally be maintained at less than 1 g/kg/day. The recommended dosing for SMOFlipid is 1–2 g/kg/day, and OO,SO-ILE formulations can be provided at doses of 1–1.5 g/kg/day. However, EFAD requirements can be met with SMOFlipid doses as low as 12.6% and OO,SO-ILE formulations as low as 12-25% of total calories. (Nutr Clin Pract 2020;35:769-82).
   iv. Biochemical evidence for EFAD (the "classic definition" is an increased triene/tetraene [eicosatrienoic acid/arachidonic acid] ratio greater than 0.2) occurs in 30%, 66%, 83%, and 100% of patients after 1, 2, 3, and 4 weeks of fat-free "full-calorie, continuous" PN (Surgery 1978;84:271-7). Clinical signs and symptoms of EFAD usually do not occur until about 2 weeks after biochemical evidence in adults. Because the investigators initiated ILE soon after the biochemical appearance of EFAD, only 2 of 32 patients developed clinical evidence suggestive of EFAD. EFAD can occur much sooner for infants and children. Patients with obesity receiving hypocaloric high-protein therapy can maintain normal plasma fatty acid profiles for up to 5 weeks (J Nutr Biochem 1994;5:243-7). Cyclic PN has been suggested to mobilize lipid from endogenous depots, but conclusive data are lacking and cyclic PN should be avoided in critically ill patients, as previously discussed.

v. Clinical symptoms (dry, scaly skin; hair loss; poor wound healing) occur about 2 weeks after biochemical evidence of deficiency in adults. Therefore, in most adults, the earliest appearance of EFAD is after about 3 weeks of fat-free full-calorie continuous PN.
vi. Serum triglyceride concentration should be monitored at least weekly and more often for those with proven or suspected impaired triglyceride clearance (consider temporarily withholding lipid emulsion when serum triglyceride approaches or exceeds 400 mg/dL) (Nutr Clin Pract 2020;35:769-82).
vii. Predisposing conditions that may result in impaired clearance of triglycerides:
(a) Excessive lipid intake (often caused by propofol therapy, which is a 10% SO emulsion containing 1.1 kcal/mL)
(b) Acute pancreatitis
(c) Uncontrolled diabetes
(d) Liver failure
(e) Kidney failure (decreased lipoprotein lipase activity, carnitine deficiency with long-term hemodialysis patients)
(f) End-stage sepsis (multisystem organ failure)
(g) History of hyperlipidemia
(h) Obesity
(i) HIV (occurred even before current antiretroviral therapy) (Am J Med 1989;86:27-31)
(j) Pregnancy
(k) Small-for-gestational-age neonates (carnitine synthesis is maturational-dependent)

e. Electrolyte requirements (see the Fluids and Electrolytes section)
f. Vitamins
i. One full dose of multiple vitamins for infusion (i.e., 10 mL/day) should be included in every bag of PN with few exceptions.
ii. Conservative supplementation doses (i.e., higher than basal needs to replace losses) of certain vitamins may be safely added to most PN admixtures (e.g. thiamine or folic acid).
iii. Repletion doses of vitamins (i.e., to correct a known deficiency) should be provided separately from the PN admixture because of numerous possible interactions affecting compatibility and stability of the admixture (JPEN J Parenter Enteral Nutr 2022;46:273-99).
g. Trace minerals
i. Zinc 3 mg/day normal requirements; 5 mg/day during critical illness; increased requirements for patients with diarrhea, intestinal fistulae. An additional 10 mg/day for a total of 13 mg/day is usually sufficient to meet increased intestinal losses (Gastroenterology 1979;76:458-67). Deficiency is characterized by loss of hair; erythematous rash, especially in periorbital regions of face; poor wound healing. Classic zinc deficiency is termed *acrodermatitis enteropathica*.
ii. Copper 0.3–0.5 mg/day is usually sufficient (Gastroenterology 1981;81:290-7). Historically, copper deficiency was considered to be a rare condition, but incidence of deficiency has become more common, particularly in patients who have undergone bariatric surgeries for weight loss and patients in the ICU who have received greater than 7–10 days of continuous renal replacement therapy (Curr Opin Crit Care 2021;27:367-77; Nutr Clin Pract 2018;33:439-46). Classic presentation of copper deficiency includes a microcytic anemia unresponsive to iron therapy, pancytopenia, neuropathies and myelopathies presenting as numbness, tingling, pain, weakness, loss of coordination, falls, vision changes, and dysphagia. Clinical signs may mimic myelodysplastic syndromes and may be masked by findings that are common among critically ill patients (Practical Gastro 2020;44:24-32). Rather than omitting copper from PN in patients with hyperbilirubinemia (a common practice historically because of concern for accumulation), this author suggests checking a serum copper level along with a c-reactive protein level with empiric provision of copper 0.3 mg daily in PN until copper level returns (typically a send-out test). Because copper levels increase in the setting of inflammation, one can be confident that a serum copper level below the normal range in the presence of an elevated C-reactive protein concentration is indicative of deficiency (Clin Nutr 2022;41:1357-424). Close monitoring of levels should continue in this case while short courses of repletion doses are provided, especially if clearance of copper is impaired (e.g. with severe hyperbilirubinemia).

iii. Chromium 10–12 mcg/day normal requirements, up to 20 mcg/day for diarrhea. Commercially available PN components have significant levels of chromium contamination. One study found that chromium was present in 65.6% of all PN components (JPEN J Parenter Enteral Nutr 2019;43:970-76). Therefore, supplementation with additional chromium in the PN formula is usually unnecessary. Deficiency is rare. Classic presentation of deficiency is hyperglycemia.

iv. Manganese 150–300 mcg/day is likely sufficient. Some studies state that the amount of manganese contamination in the compounding of PN may be adequate as opposed to supplementation. Other literature supports a dose of 55 mcg/day in PN formula to maintain stable whole blood manganese levels (Am J Clin Nutr 2002;75:112-8). Deficiency is very rare. Deficiency has been reported to present as a "diaper rash." Several case reports of manganese toxicity associated with liver disease and high manganese intake (800 mcg – 1 mg/day) (Nutrition 2001;17:689-93). Signs and symptoms of toxicity emulate those of Parkinson disease.

v. Selenium 60 mcg/day up to 120 mcg/day for patients with diarrhea or short bowel syndrome. Deficiency results in extreme muscle weakness and congestive cardiomyopathy. Classic presentation with cardiomyopathy has been termed *Keshan disease* (named after a province in China where the first cases of selenium deficiency with cardiomyopathy were discovered).

vi. Typical trace elements requirements can be provided with 1 mL per day of trace elements injection 4 (dose per 1 mL: zinc 3 mg, copper 0.3 mg, manganese 55 mcg, selenium 60 mcg). This is the first FDA-approved multi-trace product, and it replaced MTE-5, which was phased out of production in 2020. It is important to note that this four ingredient product is not equivalent to the previously available multi-trace 4 (MTE-4) product as it contains different doses of trace elements and does not contain chromium. Its formulation is more consistent with current recommendations for parenteral trace elements requirements than previous multi-trace elements products.

vii. It is common clinical practice to withhold copper and manganese in the PN formulation for patients with hepatobiliary/cholestatic liver disease or a direct (conjugated) bilirubin concentration greater than 2 mg/dL. With this practice, the multitrace elements (MTE) product is omitted from the PN solution, and zinc and selenium are added separately. However, with the MTE product reformulation in 2020, which substantially reduced the dose of copper and manganese, and the growing body of literature describing micronutrient deficiencies in critically ill patients, it is unclear whether this practice is warranted for all patients with hyperbilirubinemia (refer to previous section on copper for more information).

viii. Some clinicians withhold selenium for patients with significant renal disease who do not receive hemodialysis or CRRT, though data in support of this practice are lacking. This can be accomplished by providing the desired trace element ingredients individually.

h. Shortages of PN ingredients

i. Numerous global and national events have resulted in critical shortages of almost every PN ingredient over the last several decades. This has led to errors and substandard practice for PN management (JPEN J Parenter Enteral Nutr 2012;36:44S-7S).

ii. Considerations for management of PN shortages are published by ASPEN and should be referenced during times of shortages (www.nutritioncare.org/ProductShortages/).

6. Should supplemental PN be given to patients intolerant of EN?

a. ESPEN ICU guidelines (2023) (Clin Nutr 2023;42:1671-89): The safety and benefits of PN should be weighed on a case-by-case basis for patients who do not tolerate EN at goal. When EN is not feasible for patients at high nutrition risk or those who are severely malnourished, initiation of low-dose PN should be carefully considered and balanced against the risks of overfeeding and refeeding.

b. ESPEN EN (2006) (Clin Nutr 2006;25:210-23): For patients intolerant of EN, supplemental PN should be considered. Overfeeding should be avoided.
c. ASPEN (2022) (JPEN J Parenter Enteral Nutr 2022;46:12-41). For adult critically ill patients receiving EN, it is recommended that supplemental PN (SPN) should not be initiated prior to day 7 of ICU admission. This is a departure from the 2016 guidelines, which suggested initiation of SPN after 7–10 days if patients are not able to meet at least 60% of energy and energy requirements (JPEN J Parenter Enteral Nutr 2016;40:159-211). It should be noted that the studies evaluated in the 2022 guidelines did not include patients with malnutrition.
d. Canadian Practice Guidelines Update (2014) (Nutr Clin Pract 2014;29:29-43): It is strongly recommended that early supplemental PN or large volumes of hypertonic dextrose solutions not be used in unselected critically ill patients (i.e., low-risk patients with short stay in ICU). In the patient who is not tolerating adequate EN, data are insufficient to recommend when PN should be initiated.
e. Casaer/EPaNIC study (N Engl J Med 2011;365:506-17) mostly surgical patients, randomized controlled trial: EN only x 7 days; then PN initiated (hypertonic dextrose solutions x 2 days; then PN) versus supplemental PN in addition to whatever EN patients can tolerate during the first 7 days
    i. Worsened survival expressed as discharged alive from the ICU within 8 days (72% vs. 75%), more infections (26% vs. 23%), ICU length of stay greater than 3 days (51% vs. 48%) with early supplemental PN
    ii. The patient population was limited because malnourished (BMI less than 17 kg/m^2) patients were excluded. In addition, about 60% of the population were cardiac surgery patients, for whom the indication for PN is questionable, 50% of patients were extubated by ICU day 2, and 70% of patients had an ICU length of stay of only 3–4 days (which would imply a questionable severity of critical illness). Finally, only 58% of patients in the early PN group were even given PN (for 1–2 days), and only 25% patients in the late PN group ever received PN.
f. Heidegger (2013) (Lancet 2013;381:385-93): 307 patients, two medical centers, randomized controlled trial: Patients who received less than 60% target from EN by day 3 with an anticipated ICU stay greater than 5 days received supplemental PN or EN alone. Supplemental PN was discontinued by day 8.
    i. Supplemental PN group had decreased infections (27% vs. 38%).
    ii. Smaller study than the Casaer study. Not all patients had resting energy expenditure measured (some were predicted resting energy expenditure). Protein target was only 1.2 g/kg/day. No difference in ICU/hospital length of stay, mortality.
g. Gao (2022) (JAMA Surg 2022;157;384-93): Randomized, controlled trial of 229 patients categorized as high nutrition risk (per NRS-2002) with poor tolerance to EN after major abdominal surgery. Early supplemental PN (day 3; E-SPN) was compared with late SPN (day 8; L-SPN). Increased energy delivery and fewer nosocomial infections were observed in patients in the E-SPN group with no differences in other complications. Application to clinical practice may be limited by homogeneity of study population (primarily male, Chinese) who were relatively healthy (most patients had no comorbidities and barely met criteria for high nutrition risk with NRS-2002 score of 3 for majority of patients).
h. Summary: When to initiate supplemental PN in a critically ill patient remains controversial. The 2016 SCCM/ASPEN guidelines recommend that supplemental PN be considered in all patients unable to receive at least 60% of target EN by 7–10 days, and the 2022 ASPEN guidelines recommend against SPN initiation prior to day 7 of ICU admission. However, supporting evidence has notable limitations. Many of the patients in the Casaer study should not have been given PN in the first place as it was not indicated. The majority of those patients were discontinued from ventilator support within 2 or 3 days, able to eat, and discharged from the ICU within 4 to 5 days. This study is a reminder that PN should not be given indiscriminately to patients as supported by the VA Cooperative Trial (1991) because PN resulted in increased infectious complications in

perioperative GI cancer patients who were not malnourished as opposed to improved morbidity when given to those who were malnourished. The Heidegger data showed a benefit from supplemental PN in a sicker patient population than Casaer et al.'s population. Short-term supplemental PN may be indicated for patients who are anticipated to have prolonged duration of inability to use GI tract, anticipated prolonged duration of ICU stay, and who are either malnourished or who have a high level of catabolism. An evaluation of selected studies that support and question the use of supplemental PN was recently published (Nutr Clin Pract 2018;33:359-69).

Patient Cases

10. A 40-kg woman admitted to the trauma ICU receives a PN solution containing 350 g of dextrose, 160 g of amino acids, and 60 g of lipid daily. She has normal renal and hepatic function. Her most recent ABG from the morning shows a pH of 7.30, P_{CO_2} 55 mm Hg, P_{O_2} 96 mm Hg, and HCO_3 31 mEq/L. Her fingerstick BG values from the past 24 hours are 180–200 mg/dL. Which would be best to recommend regarding her PN?
    A. Decrease dextrose to 175 g/day, and increase lipid to 120 g/day.
    B. Add 20 units of regular human insulin per day to the PN solution.
    C. Decrease all the macronutrients by about one-half.
    D. Increase the acetate content of the PN solution.

Questions 11–12 pertain to the following case.
A 60-year-old man (weight 65 kg; about 90% IBW) is admitted for acute pancreatitis. After 4 days, he still has significant abdominal pain and distension without resolution. He has a fever, and blood/urine cultures are obtained. A CT scan of the abdomen reveals an ileus and an intra-abdominal abscess. He is given a goal PN regimen containing 300 g of dextrose, 70 g of amino acids, and 40 g of 20% lipids daily.

11. Which best depicts the kilocalories and protein this regimen will provide?
    A. 26 kcal/kg/day and 1.1 g/kg/day
    B. 26 kcal/kg/day and 1.5 g/kg/day
    C. 28 kcal/kg/day and 1.5 g/kg/day
    D. 30 kcal/kg/day and 1.5 g/kg/day

12. Which changes would be best for this patient's PN regimen?
    A. Increase dextrose to 400 g/day.
    B. Decrease dextrose to 200 g/day.
    C. Increase protein to 100 g/day.
    D. Increase lipids to 70 g/day.

E. Timing of Initiation of Nutrition Support: Early or not for ICU patients?
    1. ESPEN EN (2006) (Clin Nutr 2006;25:210-23): Questionable benefit of early versus delayed EN; however, it was recommended that critically ill patients who are hemodynamically stable and have a functioning GI tract be fed early (less than 24 hours), if possible.
    2. ESPEN ICU guidelines (2023) (Clin Nutr 2023;42:1671-89): If oral intake is not possible and there are no contraindications to EN, EN should be initiated within 48 hours rather than delaying EN or initiating PN.

3. SCCM/ASPEN (2016) (JPEN J Parenter Enteral Nutr 2016;40:159-211): Early EN should be initiated within 24 to 48 hours in the critically ill patient who is unable to maintain volitional intake once the patient is hemodynamically stable. It is suggested that very early EN (within 4 to 6 hours if possible) be initiated in a patient with burn injury.
4. Surviving Sepsis Campaign guidelines (2016) (Crit Care Med 2017;45:486-552): EN should be initiated early rather than a complete fast or only intravenous glucose in critically ill patients with sepsis or septic shock who can be fed enterally.
5. Summary: The data are confusing because several studies have used different times for the definition of early nutrition therapy: 24, 36, 48, and 72 hours. Most evidence-based clinicians would suggest that enteral nutrition therapy be initiated for most patients within 48 hours of ICU admission and no later than 72 hours. Surgical ICU patients, including those with trauma and thermal injury, have been more consistently shown to benefit from early EN as opposed to medical ICU patients, for whom results are more variable.

F. Glycemic Control
   1. Definition of the appropriate BG target range
      a. Society of Critical Care Medicine (SCCM) guidelines (2012) (Crit Care Med 2012;40:3251-76): A BG of 150 mg/dL or greater should trigger initiation of insulin therapy to keep BG less than 150 mg/dL for most patients and maintain BG absolutely less than 180 mg/dL.
      b. ASPEN guidelines (2013) (JPEN J Parenter Enteral Nutr 2013;37:23-36): A target BG range of 140–180 mg/dL is recommended.
      c. Surviving Sepsis Campaign guidelines (2016) (Crit Care Med 2017;45:486-552): An insulin dosing protocol to keep BG less than 180 mg/dL, rather than an upper target of 110 mg/dL, is recommended when the patient has two consecutive BG measurements greater than 180 mg/dL.
      d. American Diabetes Association (2023) (Diabetes Care 2023;46(suppl 1):S267-78): An insulin infusion should be used to control hyperglycemia, starting with a threshold of greater than or equal to 180 mg/dL. BG should be maintained between 140 and 180 mg/dL for most critically ill patients. More stringent goals such as 110–140 mg/dL or 100–180 mg/dL may be more appropriate for selected critically ill patients, as long as this can be accomplished without significant hypoglycemia.
      e. Summary: Many evidence-based clinicians use a target BG range of 140–180 mg/dL when caring for patients in a mixed medical-surgical ICU. A growing amount of evidence from smaller studies shows that certain subpopulations such as trauma, traumatic brain injury, cardiothoracic surgery, and thermal injury may benefit from tighter BG (e.g., less than 140–150 mg/dL) control if it can be done safely without hypoglycemia (Crit Care Med 2012;40:3251-76; Nutr Clin Pract 2014;29:534-41).
   2. BG monitoring frequency
      a. SCCM guidelines (2012) (Crit Care Med 2012;40:3251-76): BG should be monitored every 1–2 hours for most patients receiving an insulin infusion; monitoring every 4 hours is not recommended because of the risk of unrecognized hypoglycemia.
      b. Surviving Sepsis Campaign guidelines (2016) (Crit Care Med 2017;45:486-552): BG should be monitored every 1–2 hours during the insulin infusion and then extended to every 4 hours thereafter once stability in BG control is achieved.
      c. American Diabetes Association (2023) (Diabetes Care 2023;46(suppl 1):S267-78): BG should be monitored every ½–2 hours during the insulin infusion.
   3. Hypoglycemia
      a. Most guidelines define hypoglycemia as a BG less than 70 mg/dL because increased glucagon, catecholamine, and growth hormone production occurs when the BG falls below this concentration. Mild to moderate hypoglycemia is usually defined as a BG concentration of 40–60 mg/dL (because autonomic symptoms often appear) and severe (life-threatening) hypoglycemia as less than 40 mg/dL.

b. Most common risk factors for hypoglycemia during insulin therapy (Crit Care Med 2007;35: 2262-7; Crit Care Med 2006;34:96-101)
   i. Excessive insulin dose
   ii. Abrupt discontinuation of EN or PN without an adjustment in the insulin therapy (purported to cause 62% of severe hypoglycemic events in the 2006 Van den Berghe trial of medical ICU patients) (Crit Care Med 2012;40:3251-76)
   iii. Decreasing steroid dose while on insulin therapy
   iv. Hepatic failure
   v. Renal failure (half-life of insulin is prolonged, impaired renal gluconeogenesis in response to hypoglycemia)
   vi. Advanced age
   vii. Inotropes, vasopressor agents, octreotide with insulin therapy
   viii. Sepsis

4. Transitioning from a continuous intravenous insulin infusion
   a. Lack of a transition plan results in loss of glycemic control. Different methods have been described in the literature, and the best approach depends on whether the patient is transitioning to an oral diet, bolus EN, or continuous EN (JPEN J Parenter Enteral Nutr 2013;37:506-16; Crit Care Med 2012;40:3251-76; ASPEN Adult Nutrition Support Core Curriculum 2012:580-602).
   b. Once stable, ICU patients with type 1 or 2 diabetes receiving insulin infusions at greater than 0.5 unit/hour or those with stress-induced hyperglycemia receiving at rates greater than 1 unit/hour should be transitioned from an intravenous insulin infusion to a basal-bolus insulin regimen before the insulin infusion is discontinued. Several criteria should be met before transitioning to a basal-bolus regimen, including no foreseeable interruptions to nutrition for procedures, resolution of any peripheral edema, and discontinuation of all vasopressors (Crit Care Med 2012;40:3251-76).
   c. Transitioning from a continuous intravenous regular insulin infusion to a subcutaneous basal-bolus regimen: A long-acting insulin such as glargine administered every 24 hours can be used for basal coverage with initial dosing at 60%–80% of the total daily dose required from the insulin infusion. The first dose should be administered 2–4 hours before discontinuing the insulin infusion (Crit Care Med 2012;40:3251-76). Factors affecting serum blood glucose and patient response to insulin therapy should be incorporated into clinical decisions related to insulin dosing. Bolus insulin regimens are designed with rapid-acting, short-acting, or regular insulin doses to be provided as scheduled doses before meals and/or correction doses based on BG measurements (i.e., every 4–6 hours); correction doses based on BG measurements may be more appropriate for patients receiving continuous forms of nutrition support such as EN or PN. A protocol for using intermediate-acting insulin (NPH [neutral protamine Hagedorn]) has also been described for basal insulin coverage in patients receiving continuous enteral nutrition (JPEN J Parenter Enteral Nutr 2013;37:506-16).

Patient Case

13. A 55-year-old woman (weight 75 kg) without diabetes is given PN after a major GI resection. She has been weaned from mechanical ventilation and is being transferred from the ICU to the floor. Her current PN formulation is 200 g of dextrose (1.8 mg/kg/minute), 110 g of amino acids, and 80 g of lipids (1.1 g/kg/day), which meets her goal requirements at 26 kcal/kg/day and 1.5 g/kg/day of protein. It contains regular human insulin at 20 units/day. During the past 24 hours, her fingerstick BG measurements have been 170–210 mg/dL, and her serum glucose concentration is 182 mg/dL. She has received 14 units of sliding-scale regular human insulin coverage. Which would be best to suggest for optimal glycemic control?

A. Increase regular insulin to 30 units/day.
B. Decrease dextrose to 100 g/day.
C. Increase regular insulin to 50 units/day.
D. Do not change the current regimen.

G. Nutrition considerations with coronavirus disease 2019 (COVID-19)
   1. Several guidance documents have been released with specific recommendations for nutrition for patients with COVID-19 admitted to the ICU (https://www.nutritioncare.org/COVID19/). General recommendations from critical care nutrition guidelines (JPEN J Parenter Enteral Nutr 2016;40:159-211; Clin Nutr 2023;42:1671-89) are still applicable with an emphasis on preserving resources, clustering of care, and modifying clinical practices for prevention of transmission. Of note, these guidelines are primarily based on expert opinion, and many recommendations were made with the consideration of minimizing the risk to hospital staff. Publication of these recommendations also predates vaccine availability, and the guidelines may represent a risk-benefit perspective that warrants reconsideration as further knowledge is gained about the care of critically ill patients with COVID-19.
   2. Evidence suggests that patients with COVID-19 who are at higher nutrition risk have poorer outcomes (JPEN J Parenter Enteral Nutr 2021; 45:32-42.). Therefore, close monitoring of their nutritional adequacy is essential.
   3. Enteral nutrition
      a. Bedside enteral tube placement is preferred, and some guidance documents recommend that complicated feeding tube placements (involving radiology or endoscopy) should be avoided.
      b. Continuous EN is preferred over bolus EN regimens.
      c. Supplements to EN (i.e., protein or fiber) should be scheduled once daily instead of more frequent dosing.
      d. Weight-based estimations for requirements are preferred to minimize healthcare workers' exposure for indirect calorimetry. However, ongoing research suggests a persistence of hypermetabolism for up to 3 weeks after intubation in some patients with COVID-19, in which case weight-based equations may significantly underestimate energy needs (Crit Care 2020;24:581).
      e. Prone positioning is not a contraindication to EN. Early EN practices should continue in this setting, with initiation at trophic doses and advancement as tolerated. The head of the bed should be elevated to at least 10 to 25 degrees.
   4. Parenteral nutrition
      a. PN should be considered earlier if practical or clinical barriers to EN are present and prevent adequate nutrition delivery.
      b. Gastrointestinal involvement of COVID-19 may result in intolerance to EN and warrant initiation of PN.
      c. PN formulas should be formulated with the minimum volume, especially in the presence of Acute Respiratory Distress Syndrome or volume overload.
      d. Careful monitoring of serum triglycerides is warranted with provision of ILE due to reports of severe hypertriglyceridemia in patients with COVID-19.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

Hyperglycemia and other causes of non-hypotonic hyponatremia have been excluded. Urine osmolality is greater than 100 mOsm/kg, which rules out psychogenic polydipsia, and a large amount of hypotonic fluids were not being given. Urine sodium was greater than 30 mEq/L, and the patient did not receive diuretic therapy or have kidney disease. The patient appeared to be normovolemic without evidence of significant edema (expansion of the ECF compartment). Because the patient also has pneumonia (a common cause of SIADH), all of these factors indicate that the patient has hyponatremia caused by SIADH (Answer D is correct). Answer A is incorrect because the serum glucose concentration is not high enough to cause hyponatremia; nor has the patient received mannitol, glycine: nor is the patient hypertriglyceridemic. Answer B is unlikely because there was no evidence of adrenal insufficiency given in the case. Answer C is impossible because the patient did not have a traumatic brain injury.

2. **Answer: B**

Fluid restriction is the most appropriate treatment of SIADH (Answer B is correct). The "vaptans" may also be considered; however, this was not a choice. Answer A is incorrect because giving salt tabs may result in worsened fluid overload and edema. Answer C is incorrect for the same reason as Answer A. Answer D is incorrect because the severity of hyponatremia and lack of symptoms does not warrant the emergency use of hypertonic saline.

3. **Answer: D**

The best way to fluid-restrict an enterally fed patient is to use the most concentrated formulas, which are the 2-kcal/mL formulations that are specifically designed for patients with congestive heart failure. Unfortunately, protein intake may be inadequate with the use of these formulations in certain populations, and supplemental protein may have to be provided. Answers A, B, and C are incorrect because they do not result in the appropriate therapeutic decision to reduce fluid intake.

4. **Answer: B**

These dosages should be selected as the correct answer because they follow the dosing guidelines given in this chapter (unlike the doses given in answers A, C, D). Given that the patient is NPO with intolerance to anything by mouth, oral options (Answers A and D) are not practical. If the patient had a history of significant recent weight loss or if he did not respond adequately to these doses, the dose could be increased. Answer C is not correct as the dosage of phosphorus is excessive. Supplemental potassium and phosphorus would be added to the PN solution, in addition to daily intravenous doses of potassium and phosphorus.

5. **Answer: B**

Because it takes about 48 hours for serum magnesium to redistribute, the next day's serum magnesium concentration is falsely elevated. In general, it will take 4–5 days to replete this patient's magnesium deficiency (presumably caused by chronic alcohol ingestion). Thus, answer B would be the best option for this patient. Given that the patient is NPO, oral dosing options (answer A) are less desirable. Answer C is incorrect because it is too aggressive a dosage given the current serum magnesium concentration of 2.0 mg/dL (despite it being falsely elevated). Supplemental magnesium would be added to the PN solution in addition to daily doses of intravenous magnesium sulfate if he remained low or in the low-normal range or if he was also hypocalcemic (because hypomagnesemia can elicit hypocalcemia secondary to end-organ resistance to parathyroid hormone). Answer D is incorrect as the serum magnesium concentration is falsely elevated to within the normal range but has not fully redistributed yet. It would be anticipated that the concentration will decrease the following day since it takes multiple days to replenish magnesium stores.

6. **Answer: C**

Although critical illness (Answer D) and fluid resuscitation therapy may have been a factor in the development of his hypocalcemia, massive blood transfusion is the most profound cause. Citrate, added to the blood as an anticoagulant, readily binds calcium and can cause hypocalcemia. Previous studies have shown that hypocalcemia is common when patients are given more than 5 units of blood at a time (Answer C is correct). Answer A is incorrect because a low-normal serum magnesium concentration of 1.8 mg/dL is unlikely to contribute to the pathogenesis of hypocalcemia. A serum magnesium

concentration of less than or equal to 1.5 mg/dL is more likely to contribute to the pathogenesis of hypocalcemia. Answer B is incorrect because a urine output of 0.5 mL/kg/hr is not excessive.

7. **Answer: B**

A short-term intravenous infusion of 4 g of calcium gluconate (1 g = 4.6 mEq) for 4 hours has been shown to be a safe and effective therapeutic regimen for moderate to severe hypocalcemia (ionized calcium less than 1 mmol/L) (Answer B is correct). Answer A is incorrect because it is an insufficient dosage of elemental calcium. A bolus/push dose of calcium chloride (1 g = 13.6 mEq), as given in answer C, would be an effective means for treating symptomatic severe hyperkalemia, but it would be unnecessarily aggressive for treating hypocalcemia in this patient scenario. Answer D is incorrect because the moderate to severe hypocalcemia should be corrected in this post-operative trauma patient who is anemic and at high risk for bleeding complications.

8. **Answer: A**

Examination of the pH indicates that the patient has an acidemia because it is lower than 7.35. Looking at P_{CO_2} and serum bicarbonate would indicate that the primary etiology is metabolic because both are low. Calculation of the AG (145 − 118 − 18 =9) shows that no AG is present (Answers B and C are incorrect).A correction for serum albumin concentration is not needed because it is normal. The serum lactate is near the high end of the normal range but still within the normal range. This would indicate that the patient's dehydration has not become so extreme that a significant decrease in tissue perfusion was not evident (yet). Respiratory compensation appears to be intact ([1.5 x 17] + 8 = 33 vs. 34 mm Hg) on the blood gas, and it appears that the patient's history of smoking did not compromise her ability to mount a reasonable respiratory response to the acidosis. The serum chloride of 118 mEq/L indicates hyperchloremia. A non-AG hyperchloremic metabolic acidosis is common for patients with severe diarrhea. The low serum potassium and magnesium would also strongly suggest that the patient has significant diarrhea (Answer A is correct). Answer D is incorrect because the decreased P_{CO_2} is in response to the metabolic acidosis and the patient does not have a metabolic alkalosis but rather an acidosis.

9. **Answer: C**

Initial therapy with lactated Ringer solution would be the ideal choice (Answer C is correct). Treatment with 0.9% sodium chloride (Answer B) is incorrect because it would worsen the hyperchloremic acidosis. A 5% dextrose solution (Answer D) would be a poor choice because sodium/isotonic fluids are necessary to restore intravascular volume. Because the severity of the patient's acidemia is mild (pH 7.29), aggressive therapy with sodium bicarbonate is not indicated. After the first day of lactated Ringer solution to restore intravascular volume and improve pH, it would be reasonable to change to 0.45% sodium chloride to continue to restore volume depletion, depending on the patient's response to the lactated Ringer solution (e.g., restoration of normal pH, adequate urine output, resolution of tachycardia). Thus, answer A would be incorrect as the question asked for the most appropriate *initial* treatment. Of course, aggressive potassium and magnesium repletion is indicated as well. This could be done with infusions as previously discussed, and it would be reasonable to add 20 mEq of potassium chloride per liter to the 0.45% sodium chloride upon discontinuation of the lactated Ringer solution.

10. **Answer: C**

The current PN regimen provides 61 kcal/kg/day total (glucose 6.1 mg/kg/minute and lipid emulsion 1.5 g/kg/day) and protein 4 g/kg/day. The PN regimen represents gross overfeeding of this small woman and can explain her hyperglycemia and hypercapnia. Cutting all macronutrients by about one-half would result in a more reasonable regimen for this patient: 30 kcal/kg/day (glucose 3 mg/kg/minute and lipid emulsion 0.8 g/kg/day) and protein 2 g/kg/day. Because she is so small (weight 40 kg), it would be important to double-check the weight-based calculation to see whether this new regimen is appropriate to meet her caloric needs without overfeeding by calculating the BEE using the Harris-Benedict equation for women (caloric intake should not exceed 1.3–1.5 x BEE for a critically ill patient with traumatic injuries) (Answer C is correct). Although answer A reduces the respiratory quotient of the nutrient admixture, this will not solve the primary problem of overfeeding the patient because excessive calories are being provided. Answers B and D may help with the consequences of overfeeding (e.g., hyperglycemia and respiratory acidosis) but do not address the primary

problem (overfeeding) and are inappropriate management techniques.

11. **Answer: A**
Total kilocalories per day = (300 g x 3.4 kcal/g of dextrose) + (70 g x 4 kcal/g of protein) + (40 x 10 kcal/g of lipid emulsion) = 1020 glucose kcal + 280 protein kcal + 400 lipid kcal = 1700 total kcal/65 kg = 26 kcal/kg/day. Only 20% and 30% lipid emulsions are available for PN compounding. Each solution provides 10 kcal/g of intravenous lipid (unlike 9 kcal/g with oral fat) because glycerol and phospholipids are added to the emulsion. Protein intake is 70 g/65 kg = 1.1 g/kg/day (Answer A is correct). Answers B, C, and D do not represent the correct calculations as described previously.

12. **Answer: C**
This patient is moderately stressed with a normal BMI. Appropriate energy intake would be 25–30 kcal/kg/day (1625–1950 kcal/day), and protein intake would be 1.2–2 g/kg/day (78–130 g). Increasing dextrose to 400 g/day would provide a total energy of 31 kcal/kg/day. Although this glucose intake of 4.3 mg/kg/minute does not exceed 5 mg/kg/minute, the total kcal/day from all macronutrients would exceed the recommended initial range for this patient (Answer A is incorrect). Decreasing the dextrose dose to 200 g/day would provide a total energy of 21 kcal/kg/day from all macronutrients, which is below the target energy range for this patient (Answer B is incorrect). Increasing the lipid dose to 70 g/day would provide a total energy of 31 kcal/kg/day from all macronutrients, which is above the target of 25–30 kcal/kg/day (Answer D is incorrect). A protein dose of 100 g/day would provide 1.5 g/kg/day of protein. This dose is appropriate and falls within the recommended 1.2–2 g/kg/day. Increasing the protein dose to 100 g/day while maintaining dextrose at 300 g and lipid at 40 g would provide the patient with 1820 kcal or 28 kcal/kg/day. This is an appropriate protein and energy intake for this patient (Answer C is correct).

13. **Answer: A**
Because the target BG should be within 140–180 mg/dL for this surgical patient being transferred to the floor, a modest improvement in glycemic control is indicated. Thus, answer D (no change) would be incorrect. Ideally, obligatory glucose requirements should be met (e.g., about 130 g/day plus about 80–150 g/day for wound healing) to prevent the use of amino acids for gluconeogenesis. Thus, decreasing the glucose intake to 100 g/day as described in answer B is not desirable, given the mild increases in BG concentration. The easiest method to achieve glycemic control and meet caloric needs is to modestly increase the regular human insulin content in the PN solution. Because 14 units of sliding scale insulin still appears insufficient, a modest increase in insulin appears prudent. The patient is unlikely to experience hypoglycemia with the provision of insulin at 30 units/day when given 200 g of intravenous dextrose concurrently (Answer A is correct). As the stress resolves and glycemic control improves, insulin can be decreased or eliminated from the PN solution. Answer C is incorrect because it would likely provide too much insulin, based on sliding scale coverage and current BG range, and increase the patient's risk for hypoglycemia.

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**

An ileus, usually detected on radiologic examination of the lower abdomen, indicates lack of motility and presence of distention and air within the small bowel. This is usually depicted as "dilated loops of bowel." Patients cannot be fed safely or efficaciously by the enteral route during an ileus (Answer C is correct). Answer A is incorrect because a feeding tube can be placed for enteral feeding of the patient with anorexia, and PN is not indicated. Answer B is incorrect because presence or absence of bowel sounds is not an accurate marker for assessing bowel function. Answer D is incorrect because a high gastric residual volume during enteral feeding, combined with abdominal distension, bloating, emesis, or regurgitation, can often be efficaciously treated with prokinetic pharmacotherapy or advancement of the feeding tube into the small bowel with resumption of enteral feeding. In addition, the guidelines recommend against discontinuing EN for a GRV less than 500 mL in the absence of other signs of intolerance.

2. **Answer: B**

Answer B, 0.45% sodium chloride and potassium chloride 20 mEq/L, is correct given the average electrolyte composition of gastric fluid (see Table 4 regarding the electrolyte composition of GI fluids). Answers A, C, and D are incorrect because they do not as accurately replace the electrolyte content that is lost from gastric fluid output.

3. **Answer: A**

With significant diarrhea, intravenous zinc requirements from GI fluid losses during critical illness increase from the normal requirements of 3–5 mg/day. Data analyses show that most patients with increased intestinal losses can achieve a positive zinc balance on 13 mg of intravenous zinc daily (Gastroenterology 1979;76:458-67). As a result, most clinicians provide additional zinc supplementation for patients with short bowel syndrome, intestinal fistulas, or prolonged and sustained diarrhea (Answer A is correct). Answer B is incorrect because copper is an extremely rare and unlikely deficiency to occur during parenteral nutrition therapy. Answers C and D are incorrect because intractable diarrhea losses are less likely to cause a deficiency, albeit with the exception of copper (especially with extreme intestinal bypass procedures for obesity).

4. **Answer: C**

Given the severity of the patient's condition (recent seizure from severe hyponatremia) and likely diagnosis of SIADH, the immediate goal should be to achieve a serum sodium concentration of greater than 120 mEq/L by short-term infusion of 3% sodium chloride (Answer C is correct). Conivaptan (Answer D) could then be given to correct the hyponatremia, limiting the increase in serum sodium concentration to less than 10–12 mEq/L/day. Fluid restriction is imperative and is the primary overall management technique for this patient. Answer A is incorrect because a more rapid response is imperative because of the patient's seizure and severity of the condition. Answer B is incorrect because it would be a potentially life-threatening error by providing more ADH-like substance and because it is used to treat diabetes insipidus (the opposite condition of SIADH).

5. **Answer: B**

Studies show that increases in mesenteric potassium concentrations detected by potassium sensors in the splanchnic vascular bed evoke increased renal potassium excretion (feed-forward regulation of potassium homeostasis), even before regulation by aldosterone (classic feedback regulation) (Answer B is correct). The bioavailability of enteral potassium is 95%–100% in the absence of aberrations in GI motility, function, or anatomy. A primary difference between enteral and parenteral potassium is that the rate of absorption is slower with enteral potassium (Answer A is incorrect). Intravenous potassium administration can inadvertently be infused too quickly (it is acceptable to infuse potassium at 10 mEq/hour for patients without a cardiac monitor and up to 20 mEq/hour for those with a monitor). Answer C is incorrect because potassium chloride elixir or solution is an effective means for providing potassium when given intra-gastrically. It generally only causes diarrhea when given in higher doses or when administered directly into the small bowel through a feeding jejunostomy because it is a hypertonic solution. Answer D is a nonsensible answer.

6. **Answer: D**

Answer D, *additional magnesium therapy should be given daily for the next 4–5 days,* is correct because it takes 48 hours for magnesium to equilibrate after a short-term infusion, making Answer A incorrect. Although Answer B is common in clinical practice, failure to provide additional magnesium repletion in the setting of a normal serum magnesium concentration the day after a large intravenous dosage usually causes serum concentrations to decrease below the normal range again over the next 1–2 days as magnesium equilibrates. Thus, Answer B is not ideal management and is incorrect. Answer C is incorrect because hypocalcemia should autocorrect with magnesium supplementation within 48 hours of magnesium therapy; however, calcium therapy can be given concurrently, if necessary (symptomatic or ionized calcium concentration less than 1 mmol/L).

7. **Answer: D**

The current PN formula prescribed provides 3240 kcal/day (42 kcal/kg/day with a glucose infusion rate of 5.8 mg/kg/minute), which is an excessive amount of energy for this patient, and he has possible signs of overfeeding in the form of hyperglycemia. Answer D is correct because decreasing the dextrose dose to 315 g would appropriately bring the patient's energy down to 2101 kcal/day (27 kcal/kg/day with a glucose infusion rate of 2.8 mg/kg/minute). An initial energy goal of 25–30 kcal/kg/day is appropriate for this patient, and by significantly decreasing the dextrose and total energy load provided in the PN, his hyperglycemia should improve. Answer A is incorrect because the additional insulin will not address the issue of overfeeding, and sliding-scale insulin is unlikely to provide adequate glycemic control for this patient. Answer B is incorrect because temporarily adjusting the dextrose dose will not correct the issue of overfeeding in the long term. Answer C is incorrect because the total energy provided with these changes would still provide 37 kcal/kg/day and exceed the appropriate initial goal of 25–30 kcal/kg/day.

8. **Answer: B**

This patient's pH is high at 7.46 mmHg so his primary disorder is an alkalosis (Answer D is incorrect). His serum HCO_3 is 20 mEq/L (low) and P_{CO_2} is 31 mmHg (low). His high pH and low P_{CO_2} indicate that his primary disorder is a respiratory alkalosis (Answer C is incorrect). However, his laboratory values reveal an anion gap of 19 (141 − 102 − 20 = 19), which indicates that a metabolic acidosis is present, regardless of his pH or HCO_3 values (Answer A is incorrect). His delta ratio is 1.25 ([19−14] / [24−20]), which indicates that an anion gap metabolic acidosis is present with no other hidden metabolic process. Answer B is correct.